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FINANCIAL RISK ANALYSIS AND MANAGEMENT

ASSIGNMENT
ON
THE TOPIC :

**International Energy Market Connectedness and
Investment strategies: New evidence from DCC-GARCH
 R^2 decomposed connectedness measures**

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Highlights

Applied DCC-GARCH R^2 decomposition to measure dynamic connectedness among seven international energy commodities.

Included both renewable (solar, wind, geothermal, biofuel, fuel cell) and fossil fuel markets (natural gas, crude oil).

Identified renewable energy commodities as consistent net volatility receivers and fossil fuels as net transmitters.

Demonstrates increased market connectedness during global economic and energy crises, highlighting time varying spillovers.

Developed connectedness driven portfolio strategies yielding superior hedging effectiveness and enhanced Sharpe ratios.

Provided actionable insights for energy market investors and policymakers focused on transition risks and diversification.

Abstract

This study examines the dynamic connectedness and volatility spillovers among seven key international energy commodities i.e solar energy, wind energy, geothermal energy, natural gas, crude oil, biofuel, and fuel cell , applying a novel DCC-GARCH framework combined with R^2 decomposed connectedness measures. Using daily return data and advanced multivariate volatility modeling, we uncover the time varying interdependencies and directional spillovers across renewable and traditional energy markets. Our analysis highlights distinct roles of renewable energies as net receivers and fossil fuels as net transmitters of volatility shocks, especially pronounced during periods of market stress. The study further provides insights into optimal hedging and portfolio strategies, demonstrating improved risk adjusted performance via connectedness based diversification. Our findings contribute to a nuanced understanding of energy market integration, offering implications for investors, policymakers, and risk managers facing the global energy transition.

Keywords :

Energy Markets, Renewable Energy, Volatility Spillover, Connectedness, DCC-GARCH, R^2 Decomposition, Portfolio Optimization, Risk Management, Renewable Fossil Fuel Interaction, Dynamic Conditional Correlation

1. Introduction

There have been significant changes in the global energy system driven by two factors: Energy Transition and Financial Market integration. Growing concern about climate change, decarbonization policies, and rapid technological advancements has led to an increase in clean energy investments. This has strengthened the connection between traditional and renewable energy markets, affecting risk propagation and its impact on policymakers. Understanding these interdependencies is essential for designing effective strategies for investment and portfolio diversification.

Since the 2007 financial crisis, cross market connections have changed massively. New methods have been introduced for a better understanding of volatility spillovers, market risks, and tail dependence across different asset classes. More of the market is recognizing energy markets as central to the global financial network because of their role in policymaking, pricing, and other international links. Clean energy markets are also becoming a source of volatility transmission, similar to how oil and gas markets spread systemic risks. Therefore, studying these markets is increasingly important.

The integration of energy markets into finance has created complex mechanisms that push market boundaries and generate new channels for risk transmission and price discovery. Connections between different renewable sources, market rules, and real time data have resulted in heavily interconnected energy systems. This development has led to cross border energy trading and resource allocation. However, it has also introduced some challenges, as connected networks can allow disruptions in one area of the grid to cascade to others, which can increase systemic risk.

A key driver in the recent expansion of renewable energy is financial innovation. It facilitates larger scale, higher growth clean energy investment, which spurs further expansion in the industry. In this context, the movement of energy related prices, particularly electricity prices, have begun to increasingly exhibit patterns analogous to traditional financial asset classes, including equities, commodities, and currencies. As a result, conventional hedging relationships have broken down, and firms are placing a greater emphasis on portfolio diversification strategies. Energy markets in constant change offer both challenges and opportunities for market participants and regulators alike. In the case of market integration, a fixed perspective will gloss over the full advantages of these trends and also not take into account the expected impacts and views of different investors. Understanding these interactions is essential for creating effective risk management strategies and frameworks. Econometric tools like variance decomposition and VAR (Value at Risk) models help measure spillovers. However, they have limitations, such as capturing time varying dynamics and higher order correlations. DCC GARCH R^2 provides a new way to simplify the total connectedness across various markets. It helps us understand how systemic risk develops and spreads under different market conditions. Investors can use this information for portfolio allocation.

This study looks at the international energy market and its implications for investment strategies and its impact on financial markets using the DCC GARCH R^2 decomposed connectedness approach. Considering recent contributions in the literature on tail risk contagion, volatility spillovers, and the links between clean and fossil energy, we broaden the analysis to include a wider range of energy commodities, indices, and other related asset classes from different economies. The paper makes three key contributions. First, it provides us evidence on the structure and connectedness among international energy markets. Second, it offers insights for international investors by quantifying the benefits and losses that arise from changes in systemic linkages. Third, it presents lessons relevant to policy on integrating energy finance.

2. Literature Review

Gong et al. (2023) examines tail risk spillovers between international energy markets, including both traditional and clean energy, which uses a quantile vector autoregression (QVAR) model from 2010 to 2022. They found out that during extreme market conditions, spillovers are significantly stronger, particularly in the long run. Traditional energy acts as the main transmitter, while many clean energy types, such as solar, geothermal, and fuel cell, are net receivers. This indicates that relying solely on average or normal shock connectedness overlooks major risks during market crashes.

Cocca et al. (2024) analyze four NASDAQ clean energy indices—Green Economy, Solar, Wind, and Geothermal—from December 2010 to June 2023, using a DCC-GARCH model with R^2 decomposition. They found that connectedness among these indices changes over time and in response to economic events. The Green Economy index frequently sends out shocks, while the others mainly receive them. Their portfolio strategies based on this analysis outperform the Green Economy index and other options in terms of risk management and return, measured by the Sharpe ratio.

Foglia et al. (2022) study firm level tail risk spillovers among 32 clean energy firms and 8 oil firms from January 2011 to October 2021 using the TENET (Tail Event driven NETwork) model. They observe that most risk spillovers occur within each sector, with oil firms influencing other oil firms and clean energy firms affecting each other. Weak spillovers happen across sectors. Some clean energy firms transmit tail risk, while oil firms are more commonly net receivers. This emphasizes the importance of sector clustering and firm differences when considering extreme risk.

Pham (2020) studies the connection between clean and dirty energy investments during extreme returns. He uses a quantile VAR approach with global data. He found out that the returns of clean and dirty energy are closely linked during market stress. The spillovers become uneven in times of crisis. Key drivers of this extreme return connection include macro uncertainty, volatility, and investor sentiment.

Zhang et al. (2020) look at the time and frequency dynamics of the relationship between renewable energy stocks and crude oil prices. They use wavelet and VAR models. Their results indicate that short-term spillovers dominate during periods of high volatility. The Long-term spillovers showed structural changes, like energy transitions. Renewable energy returns respond more to oil shocks than the other way around.

Baruník and Křehlík (2018) create a method for analyzing connectedness based on frequency dynamics, applying it to financial and commodity markets. They showed that the connectedness varies by frequency: short-term spillovers correspond to crises, while long-term spillovers reflect macro trends. This framework is widely used in later studies of energy markets.

Diebold and Yilmaz (2012) introduce the predictive directional spillover index for volatility connections. This is based on forecast error variance decompositions. They demonstrate that shocks in one market can predict changes in volatility in others. Their method lays the foundation for many subsequent studies on connectedness in finance and energy.

Rehman et al. (2022) study the changing volatility spillovers among commodity markets during COVID-19 using a DCC-GARCH connectedness approach. They found that spillovers increase

significantly during the pandemic, with energy commodities driving the shocks. This emphasizes the need for considering changing connectedness in risk management during crises.

Li et al. (2024) analyze tail risk contagion and connectedness between crude oil, natural gas, heating oil, precious metals, and international stock markets using a quantile VAR model. They found strong spillovers across markets, especially from oil and natural gas to stocks and metals. The contagion effects are most evident during geopolitical shocks and market crashes.

Wu et al. (2022) investigate connectedness in global energy markets using time-frequency methods on fossil and clean energy prices. They showed that spillovers are stronger in the short term, especially during oil shocks and renewable policy changes. In the long term, clean energy is becoming more integrated with fossil markets, which reduces diversification benefits.

Gupta et al. (2024) examine dynamic volatility connectedness between fossil energy, clean energy, and major asset classes using a VAR-BEKK-GARCH framework. They found significant bidirectional volatility spillovers, with fossil energy often leading to clean energy. Financial assets like equities and bonds also send shocks into energy markets, making cross-market risk management crucial.

Jiménez-Rodríguez (2020) reviews volatility spillovers in energy markets in a doctoral thesis, comparing GARCH, VAR, and frequency-domain approaches. He found that oil consistently transmits shocks, while gas and renewables usually receive them. The choice of method affects spillover estimates, which highlights the need for robustness checks.

Chen et al. (2025) study intraday extreme risk spillovers in the energy sector using high-frequency data and a quantile spillover approach. They observe strong contagion effects within a day, with oil and gas transmitting shocks to clean energy. They also showed that portfolio strategies that account for intraday spillovers improve hedging effectiveness.

Wang et al. (2024) analyze tail risk connectedness during geopolitical shocks using quantile connectedness methods. They found that events like the Russia-Ukraine war sharply increase tail spillovers across energy and stock markets. Oil and gas are the main transmitters, while the role of clean energy as a receiver strengthens during crises.

Xu et al. (2019) measure tail risk in crude oil futures using GAS time-varying copula models with fat-tailed GARCH distributions. They found that tail dependence between oil and other assets is stronger during turbulent markets. The results indicate that overlooking fat tails leads to an underestimation of risk exposure in commodity-linked portfolios.

Henriques and Sadorsky (2015) explore dependence and systemic risk between oil and renewable energy stocks using VAR and Granger causality methods. They discover that oil price shocks greatly impact renewable energy stock prices, but the reverse effect is weak. Renewable energy remains vulnerable to oil market volatility, limiting its role in diversification.

Bouri et al. (2023) investigate dynamic connectedness among dirty and clean energy markets using a TVP-VAR model. They found that connectedness increases during crises like COVID-19 and during discussions about the energy transition. Clean energy mostly acts as a net receiver, while fossil energy transmits shocks across the system.

Zhao et al. (2024) examine the connection between crude oil and clean energy stock indices using TVP-VAR and CAViaR models. Their results showed significant tail spillovers from oil to clean

energy, especially during volatile periods. Risk spillovers are nonlinear and asymmetric, emphasizing the role of extreme events in shaping the oil-renewables relationship.

Chen et al. (2023) study connectedness and risk spillovers between crude oil and clean energy stock returns using quantile Granger predictability tests. They found that oil strongly predicts clean energy returns in extreme quantiles, while clean energy has limited predictive power for oil. Tail risks dominate the connectedness structure.

Zhang and Liu (2025) analyze tail risk spillovers between global stock markets using methods based on entropy. They found significant global tail contagion, with energy-related stock markets being among the most affected. Spillovers increase during crises, which reduces the benefits of international diversification.

Hu et al. (2025) look at tail risk connectedness between energy tokens (crypto-assets linked to energy) and FinTech stocks using a quantile spillover approach. They showed that tail spillovers are strong and asymmetric, with FinTech often sending shocks into energy tokens. This highlights new systemic risks in digital asset markets.

Ando et al. (2017) suggest a large t-VAR framework to examine volatility spillovers with heavy tails. When applied to financial markets, they showed that ignoring fat tails leads to an underestimation of systemic risk. Their approach captures extreme connectedness better, making it useful for energy-finance applications.

Gao et al. (2020) study connectedness in clean energy stocks using a frequency-domain VAR model. They found that clean energy sectors are highly interconnected, especially in the short term. Oil shocks affect renewables mainly through the long-term frequency, highlighting the multi-scale nature of connectedness.

Sinha (2025) reviews volatility spillover methods in an SSRN paper, comparing GARCH, VAR, frequency-domain, and copula-based models. He points out that time-varying and frequency-based models capture spillovers more effectively than static methods. The review offers guidance for choosing methods in energy-finance research.

Kumar et al. (2025) investigate the impact of external uncertainties on extreme return connectedness between food, fossil energy, and clean energy markets using a quantile VAR model. They found that geopolitical risks and policy uncertainty increase tail spillovers, especially from fossil energy into food and clean energy. The findings emphasize the systemic role of external shocks.

Ali et al. (2025) examine the connectedness of ESG and sustainable development using DCC-GARCH with R^2 decomposition. They found that ESG indices are linked with energy markets, with clean energy indices increasingly transmitting shocks. Investment strategies based on ESG-connectedness improve hedging and risk-adjusted performance.

Zhang et al. (2025) study time-frequency and quantile connectedness among geopolitical risks, conventional energy, clean energy, electricity, and carbon markets. They found strong multi-scale connectedness, with conventional energy leading most markets. Clean energy becomes more systemically important over time, especially in response to geopolitical shocks.

In summary, the literature highlights the central role of energy markets in global financial connectedness and the growing importance of clean energy in shaping risk transmission. Prior studies emphasize that fossil energy assets remain dominant transmitters of shocks, while clean energy indices and firms usually act as receivers, particularly during extreme market conditions. The evidence also shows that spillovers vary by time and frequency, increasing during crises and reflecting long-term structural changes. While previous research has explored connectedness among fossil and clean energy, metals, and equities, it has often focused on limited sets of assets or specific methods. This study uniquely applies the DCC-GARCH R^2 decomposed connectedness framework across a wide range of international energy markets and related asset classes, offering new insights into systemic linkages and their implications for investment strategies.

3. Research Gaps in Existing Literature

1) None of the existing studies combine extreme tail connectedness with frequency (short, medium and long-run) decomposition in a single framework.

While tail-based studies have shown spillovers are strongest in crises, and frequency studies show how shocks behave differently across horizons, we still do not know whether extreme spillovers are short lived or persistent, a critical insight for risk management and investment strategies.

2) Most existing studies on clean (CE) and dirty energy (DE) markets focus on how shocks move between them, especially during crises like the COVID-19 pandemic or the Russian-Ukrainian War. For example, Dynamic Connectedness among Dirty and Clean Energy Markets: A TVP-VAR Approach shows that CE markets now strongly influence fossil fuel markets, and that overall connectedness spikes during major events.

However, these studies mostly use TVP-VAR or quantile-based methods. They have not compared their results with the DCC-GARCH R^2 decomposed connectedness approach, which could offer different and possibly better insights. We still don't know which method gives clearer signals for building investment strategies or reducing risk during crises.

3) Despite detailed analyses of tail-risk and connectedness, existing studies mostly focus on specific asset classes or regions (e.g., clean energy vs. oil, global stock indices, energy tokens vs. FinTech). Cross-asset and cross-regional interactions under extreme market events remain underexplored. Additionally, the role of dynamic investment horizons in shaping hedging and diversification strategies could be further investigated, especially using integrated portfolios across traditional and digital assets. Finally, most studies emphasize statistical relationships without testing actionable real-world portfolio applications under varying market crises.

4) None of the studies combine the strengths of all three approaches into a unified framework. Fuentes and Herrera (2022) provide detailed sectoral insights but limit themselves to equities and implied volatility indices. Barbaglia et al. (2020) handle fat tails and high-dimensional systems, but they do not integrate DCC-GARCH or frequency decompositions. Jovović and Popović (2023) examine bonds and energy with dynamic correlations and horizon analysis, but without heavy-tailed estimation or large cross-asset coverage.

What is missing is a study that brings these elements together: a DCC-GARCH-based framework linked with R^2 or frequency-decomposed connectedness measures, while also accommodating heavy tails and sparsity for a large set of assets including equities, commodities, green bonds, and conventional bonds. Addressing this gap would provide a more complete and realistic understanding of international energy market connectedness and its implications for investment strategies.

4. Objectives of the Study

The primary objective of this study is to examine the systemic linkages and risk transmission dynamics among international energy markets utilizing advanced DCC-GARCH R^2 decomposed connectedness measures and to provide actionable insights for financial market participants and policymakers, as outlined in recent methodological literature. The study will expand the analysis beyond previous literature to a broader set of energy commodities, indices, and related asset classes from various economies. The analysis aims to assess the dynamic transmission of shocks and volatility spillovers across key energy commodities, countries, and financial asset classes, building on and updating prior findings with data spanning from January 2000 to June 2025, subject to data availability.

A secondary objective is to investigate the implications of interconnectedness for international investment strategies and portfolio diversification. By identifying transmission channels and key market influencers, the study seeks to provide actionable insights for investors and policymakers concerning risk management, hedging practices, and policy design related to the global energy-finance nexus.

The third objective is to examine the dynamics of risk transmission and volatility spillovers between conventional (fossil) and renewable energy markets in depth. Building on previous research, the study verifies if, particularly in times of crisis and harsh market conditions, clean energy indices are primarily net receivers and fossil energy assets continue to be dominating shock transmitters. It does, however, also evaluate the growing systemic significance of clean energy, which is widely acknowledged as being essential to the global financial system. The research takes into consideration the fact that spillovers vary in frequency and time, becoming more intense during crises and reflecting long-term trends and structural changes. The project specifically examines the understudied behavior of cross-asset and cross-regional relationships during extreme market occurrences. The fourth objective is to convert these connection metrics into workable investment and hedging methods. The research aims to assist investors benefit from these dynamics for portfolio allocation, diversification, and enhanced risk-adjusted performance by advancing our understanding of how systemic risk changes over time. By contrasting the outcomes of the DCC-GARCH R^2 decomposed connectivity technique with those derived from TVP-VAR or quantile-based approaches, which are widely used in the literature, the study also seeks to fill in current research gaps. Given the present literature's lack of clarity regarding whether technique provides clearer signals for appropriate investment strategies and risk reduction during crises, this comparative perspective is crucial.

5. Data and Preliminary Analysis

This study employs a daily panel constructed from the provided closing price series for seven markets, Solar (GRNSOLAR), Wind (GRNWIND), Geothermal (GRNGEO), Fuel-cell/clean-fuel (GRNFUEL), Biofuels (GRNBIO), Crude oil front-month (CL1 Comdty), and Natural gas front-month (NG1 Comdty), which were taken from Bloomberg and merged on a common trading calendar and transformed into daily log returns to form the multivariate input for the connectedness analysis. To ensure modelability and comparability across assets, prices are first sorted and deduplicated by date, dates common to all series are retained, and daily percentage log returns $rt=100 \times (\ln Pt - \ln Pt-1)$ are computed for each asset before stacking into a synchronized return matrix Y with assets SOL, WIN, GEO, FUEL, BIO, OIL, and NGAS.

Figure 1 plots the daily percentage returns for each of the seven energy assets over the sample period. Visually, all time series clearly exhibit volatility clustering, a well-known financial market phenomenon where periods of high volatility (large price swings) are grouped together, followed by periods of relative calm (smaller price swings). A particularly dramatic spike in volatility is evident across all assets around 2020, signaling a major market-wide shock. During this crisis period, the conventional energy commodities, most notably Crude Oil (CL1 Comdty), display exceptionally large negative price movements. Consistent with this observation, the plots for conventional energy (Crude Oil, Natural Gas) generally show fluctuations of a much larger magnitude throughout the sample, when compared to the five clean energy indices, which appear comparatively more stable.

Figure 1: Percentage changes

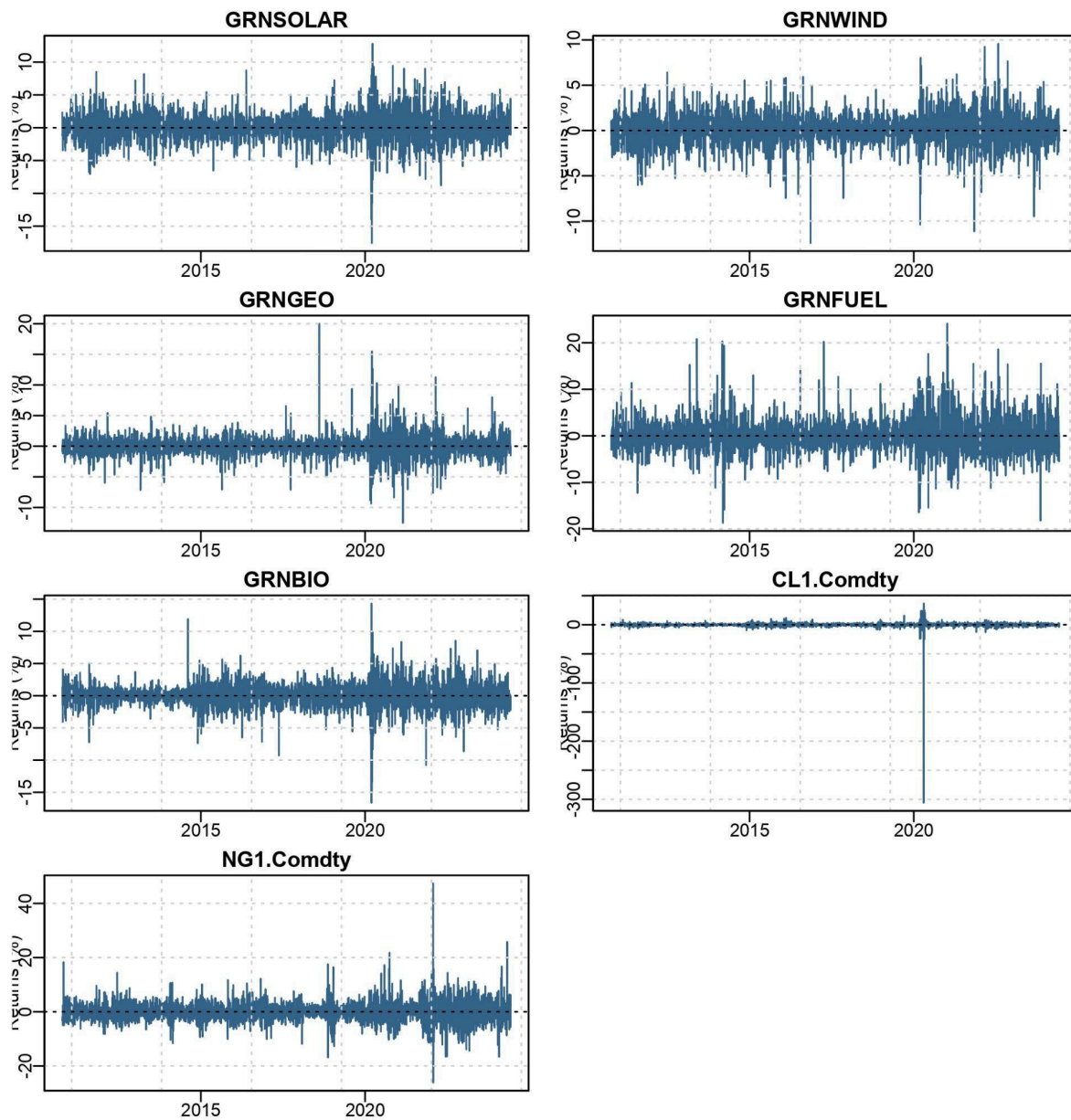


Table 1: Summary statistics

	Solar Energy	Wind Energy	Geothermal Energy	Fuel Cell	Bio Fuel	Crude Oil	Natural Gas
Mean	0.066*	0.040	0.020	0.062	0.006	-0.059	0.076
	-0.063	(0.172)	(0.475)	(0.292)	(0.858)	(0.580)	(0.219)
Variance	4.303	2.911	2.823	11.95	3.42	38.837	13.015
Skewness	-0.156***	-0.182***	0.703***	0.699***	-0.484***	-36.779***	0.821***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Ex.Kurtosis	4.408***	3.842***	13.661***	4.975***	8.666***	1737.942***	12.386***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
JB	2797.185***	2133.491***	27018.021***	3825.973***	10893.024***	433453464.912***	22364.259***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ERS	-9.615***	-15.862***	-25.315***	-25.913***	-13.112***	-22.243***	-15.624***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Q(20)	57.040***	34.918***	25.648***	22.085***	59.798***	378.173***	46.849***
	(0.000)	(0.000)	(0.001)	(0.007)	(0.000)	(0.000)	(0.000)
Q2(20)	1682.646***	241.795***	346.096***	500.638***	2822.416***	99.959***	307.950***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
kendall	Solar Energy	Wind Energy	Geothermal Energy	Fuel Cell	Bio Fuel	Crude Oil	Natural Gas
Solar Energy	1.000***	0.269***	0.246***	0.313***	0.267***	0.119***	0.010
Wind Energy	0.269***	1.000***	0.192***	0.224***	0.195***	0.058***	-0.009
Geothermal Energy	0.246***	0.192***	1.000***	0.208***	0.216***	0.091***	-0.004
Fuel Cell	0.313***	0.224***	0.208***	1.000***	0.247***	0.115***	0.025**
Bio Fuel	0.267***	0.195***	0.216***	0.247***	1.000***	0.201***	0.019
Crude Oil	0.119***	0.058***	0.091***	0.115***	0.201***	1.000***	0.082***
Natural Gas	0.010	-0.009	-0.004	0.025**	0.019	0.082***	1.000***

Table 1 presents the summary statistics for the daily percentage returns of the five clean energy indices (Solar Energy, Wind Energy, Geothermal Energy, Fuel Cell, Bio Fuel) and two conventional energy commodities (Crude Oil, Natural Gas). The significant Elliott, Rothenberg & Stock (ERS) unit-root test statistics for all series (Elliott et al., 1996) confirm their stationarity, a prerequisite for the analysis. Based on the standard t-test, only the mean return for Solar Energy (0.066) is positive and statistically significant (at the 10% level). The conventional energy commodities, Crude Oil and Natural Gas, exhibit by far the highest variance (38.837 and 13.015, respectively), indicating they are considerably more volatile and risky than the clean energy indices; among the latter, Geothermal Energy is the least volatile (2.823). All series demonstrate statistically significant skewness; Solar Energy, Wind Energy, Bio Fuel, and Crude Oil are negatively (left) skewed,

suggesting a higher probability of extreme negative returns, whereas Geothermal Energy, Fuel Cell, and Natural Gas are positively (right) skewed. Moreover, all series are found to be significantly leptokurtic (positive excess kurtosis) and non-normally distributed, as confirmed by the highly significant Jarque-Bera (JB) test statistics (Jarque & Bera, 1987), indicating fatter tails compared to a normal distribution. Finally, the significant Ljung-Box Q-statistics for squared returns (Q2(20)) (Ljung & Box, 1978) for all assets reveal the presence of strong ARCH effects, justifying the subsequent use of GARCH-family models.

A nonparametric dependence snapshot using Kendall's rank correlation reveals low-to-moderate positive association within the renewable complex (for example, SOL-FUEL ≈ 0.313 and SOL-WIN ≈ 0.269), consistent with shared but time-varying drivers. Oil's monotonic association with renewables is modest (for example, OIL-BIO ≈ 0.201 and OIL-SOL ≈ 0.118), while natural gas exhibits weak links to all others (for example, NGAS-WIN ≈ -0.009 and NGAS-SOL ≈ 0.010), motivating a dynamic connectedness framework rather than assuming stable high correlations. Taken together, the evidence of stationarity, heavy tails, serial correlation in returns, and strong ARCH effects justifies a two-stage approach in which per-asset GARCH models with fat-tailed errors generate standardized innovations for DCC correlation estimation, followed by dynamic conditional R^2 and its decomposition to quantify total and directional connectedness among the seven energy markets.

6. METHODOLOGY

6.1. Overview

This study quantifies contemporaneous interdependencies and risk transmission in international energy markets using a DCC-GARCH R-squared decomposed connectedness framework, which builds dynamic goodness-of-fit measures and decomposes them into per-asset contributions from conditional moments. The approach starts from a multivariate linear regression foundation, generalizes it via DCC-GARCH conditional variance-covariances, and stacks decomposed R-squared measures into a connectedness matrix that yields TO, FROM, NET, TCI, and pairwise indices for market-risk interpretation and portfolio design.

6.2. Data and transformations

Daily or higher-frequency prices for international energy indices or assets are transformed to percentage returns to ensure stationarity, as unit-root tests on levels typically motivate modeling in returns for volatility and correlation dynamics. Descriptive diagnostics usually reveal fat tails, skewness, and ARCH/GARCH effects in energy returns, justifying the use of flexible GARCH-family innovations and heavy-tailed distributions in the volatility stage.

6.3. MLR foundation and static R-squared

Let $Z = [y, X]$ collect the left-hand-side (LHS) return y and K right-hand-side (RHS) returns X , demeaned so no intercept is needed, with the regression $y = X\beta + \varepsilon$. Partition the unconditional variance-covariance H and correlation R as $H = \begin{bmatrix} H_{yy} & H_{xy} & H_{yx} & H_{xx} \end{bmatrix}$, and $R = \begin{bmatrix} 1 & R_{xy} & R_{yx} & R_{xx} \end{bmatrix}$ to compute coefficients and goodness of fit from moments.

The regression coefficients and R-squared follow as $\hat{b} = (X'X)^{-1} X'y = H_{xx}^{-1} H_{xy}$ and $R^2 = 1 - \frac{(y-X\hat{b})'(y-X\hat{b})}{y'y} = H_{xy}' H_{xx}^{-1} H_{xy} H_{yy}^{-1} = R_{xy}' R_{xx}^{-1} R_{xy}$.

6.4. DCC-GARCH specification and estimation

Define returns z_t with conditional variance-covariance H_t and correlations R_t , standardize

$u_t = D_t^{-1} z_t \sim N(0, I)$, where $u_t \sim N(0, I)$, $D_t = \text{diag}\left(\left\{h_{1t}^{1/2}, \dots, h_{Kt}^{1/2}\right\}\right)$, and evolve the correlation driver Q_t via $Q_t = (1 - a - b) \bar{Q} + a u_{t-1}' u_{t-1} + b Q_{t-1}$. Construct $R_t \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$ and

$H_t = D_t R_t D_t'$, estimating in two steps: fit univariate GARCH to obtain D_t , then maximize the correlation likelihood for (a, b) to obtain R_t and H_t . The log-likelihood splits into volatility and correlation parts, enabling flexible univariate model selection and robust correlation dynamics under the two-step estimator by Engle (2002).

- Summary - DCC system:

$$u_t = D_t^{-1} z_t \sim N(0, I) \text{ where } u_t \sim N(0, I)$$

$$D_t = \text{diag}\left(\left\{h_{1t}^{1/2}, \dots, h_{Kt}^{1/2}\right\}\right) Q_t = (1 - a - b) \bar{Q} + a u_{t-1}' u_{t-1} + b Q_{t-1}$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

$$H_t = D_t R_t D_t'$$

- Log-likelihood: $L(\theta, \phi) = L_V(\theta) + L_C(\theta, \phi)$ with explicit forms for volatility and correlation components
- Univariate omnibus GARCH family: $h_t^{\lambda/2} = \omega + \alpha h_{t-1}^{\lambda/2} (|z_{t-1} - \eta| - \gamma(z_{t-1} - \zeta))^\delta + \beta h_{t-1}^{\lambda/2}$ to nest GARCH/EGARCH/TGARCH/APARCH and others.

6.5. Dynamic conditional R-squared

Replace unconditional moments in the static R-squared formula with DCC-GARCH conditional moments to obtain the time-varying goodness-of-fit R_t^2 , which measures contemporaneous explanatory power at each t. Using conditional correlation blocks, the dynamic conditional R-squared is $R_t^2 = R_{xy,t}' R_{xx,t}^{-1} R_{xy,t}$, guiding interpretation of predictability, co-movement risk, and hedging feasibility in international energy markets.

To attribute multivariate explanatory power to correlated RHS assets, decompose the RHS correlation block via eigen-decomposition, transform to orthogonal latent factors, and map back to obtain a nonnegative vector of per-asset contributions summing to R_t^2 . Eigen and mapping:

$$\text{Let } R_{xx,t} = V_t \Lambda_t^2 V_t' \\ \text{and } R_{xf,t} = V_t \Lambda_t V_t' \text{ decomposed vector: } R_t^2 = R_{xf,t}^2 R_{fy,t}^2$$

6.6. Connectedness measures

Stack per-asset R_t^2 decompositions across choices of LHS to form an asymmetric R-squared connectedness matrix R_t^{2d} , where off-diagonals quantify the predictive contribution of column assets to row assets at time t . Directional FROM equals the row-sum (the multivariate R_t^2 for that LHS), directional TO equals the column-sum, NET = TO – FROM classifies transmitters vs receivers, and the Total Connectedness Index equals the cross-sectional average of conditional R_t^2 . Furthermore, the net pairwise directional connectedness illustrates the net transmission on the bilateral level. The $NPDC_{ij,t}$ represents the difference in the predictive contribution between variable i and variable j in time t . The total connectedness index (TCI) which assesses the interdependence within the network and is often used as a proxy for market risk. As the TCI is considered to be equal to the average total directional connectedness TO others or total directional connectedness FROM others, the TCI becomes equivalent to the averaged conditional R^2 goodness-of-fit measure

- Directional FROM: $FROM_{i,t} = \sum_{k=1, k \neq i}^K R_{ik,t}^{2d} = R_{i,t}^2$
- Directional TO: $TO_{i,t} = \sum_{k=1, k \neq i}^K R_{ki,t}^{2d}$
- NET: $NET_{i,t} = TO_{i,t} - FROM_{i,t}$
- Pairwise net: $NPDC_{ij,t} = R_{ij,t}^{2d} - R_{ji,t}^{2d}$
- Total Connectedness Index: $TCI_t = \frac{1}{K} \sum_{k=1}^K TO_{k,t} = \frac{1}{K} \sum_{k=1}^K FROM_{k,t} = \frac{1}{K} \sum_{k=1}^K R_{k,t}^2$
- Pairwise connectedness indices: $PCI_{ij,t}^C = 2 \cdot \left(\frac{R_{ij,t}^{2d} + R_{ji,t}^{2d}}{R_{ii,t}^{2d} + R_{ij,t}^{2d} + R_{ij,t}^{2d} + R_{jj,t}^{2d}} \right)$ based on decomposed

R-squared and $PCI_{ij,t}^R = \frac{2 \cdot R_{ij,t}^2}{1 + R_{ij,t}^2}$ for correlation-only, used in bilateral minimization strategies.

6.7. Portfolio construction for international energy

Hedging and allocation are built directly on DCC outputs: bivariate hedge ratios use : $HR_{ij,t} = \frac{H_{ij,t}}{H_{jj,t}}$

while multivariate hedging uses dynamic betas: $b_t = H_{xx,t}^{-1} H_{xy,t}$.

to short multiple assets against a long position. Optimal two-asset no-shorting weights are

$$w_{ij,t} = \frac{H_{jj,t} - H_{ij,t}}{H_{ii,t} - 2H_{ij,t} + H_{jj,t}} \quad \text{with weight constraints as} \\ w_{ij,t} = \begin{cases} 0 & \text{if } w_{ij,t} < 0 \\ w_{ij,t} & \text{if } 0 \leq w_{ij,t} \leq 1 \\ 1 & \text{if } w_{ij,t} > 1 \end{cases}$$

The global minimum risk portfolio can be seen as a minimization optimization problem which can be formulated as follows, $\underset{w_t}{\operatorname{argmin}} w_t' P_t w_t \text{ s.t. } w_t' 1 = 1, 0 \leq w_t \leq 1$.

Portfolio evaluation uses the Sharpe ratio: $SR = \frac{\bar{x}_p}{\sqrt{\operatorname{var}(x_p)}}$

and hedging effectiveness: $HE_i = 1 - \frac{\operatorname{var}(x_p)}{\operatorname{var}(x_i)}$

with VaR- and CVaR-based alternatives for downside-focused ratios. These metrics assess whether minimizing connectedness or correlation improves reward-to-risk and risk reduction relative to international energy benchmarks and constituent assets.

7. Empirical Results

7.1. DCC-GARCH estimation

Table 2 presents the essential diagnostic tests for the univariate GARCH models fitted to each energy asset, which serve as the foundation for the subsequent multivariate DCC-GARCH analysis. The 'SignBias' test checks for asymmetric volatility responses not captured by the symmetric GARCH model. For all seven assets, the p-values are high and therefore statistically insignificant (e.g., Solar Energy: 0.593, Bio Fuel: 0.897), indicating that the null hypothesis of no sign bias cannot be rejected. This confirms that the symmetric GARCH specification is adequate. Second, the 'WARCH(20)' test, a weighted ARCH-LM test (Engle, 1982), checks for any remaining ARCH effects (autocorrelation) in the squared standardized residuals. The models perform exceptionally well, with high p-values for nearly all assets (e.g., Wind Energy: 0.958, Crude Oil: 1.000), demonstrating that the GARCH models successfully capture volatility clustering. The only minor exception is Solar Energy, which, with a p-value of 0.0489, shows slight evidence of remaining ARCH effects at the 5% level, though it is still considered sufficiently well-specified to proceed.

The latter half of the table evaluates the practical risk management performance of the models by backtesting their 1% Value-at-Risk (VaR) forecasts. The 'VaR' column reports the p-value for the unconditional coverage test (Kupiec, 1995), which assesses whether the frequency of VaR exceptions aligns with the expected 1%. The high p-values for all series (e.g., Geothermal Energy: 0.759, Fuel Cell: 0.931) confirm the accuracy of the VaR models. Furthermore, the 'VaR Dur.' test (Christoffersen & Pelletier, 2004) checks for the independence of VaR exceptions, a crucial requirement. Once again, the high p-values (e.g., Wind Energy: 0.387, Crude Oil: 0.675) indicate that the VaR violations do not cluster, satisfying the independence assumption. Coupled with the robust p-values for the Conditional Value-at-Risk (CVaR) backtest, these results strongly validate the univariate GARCH models as being well-specified and reliable for capturing the dynamic risk profile of each energy asset.

Table 2 : Evaluation of univariate GARCH performance

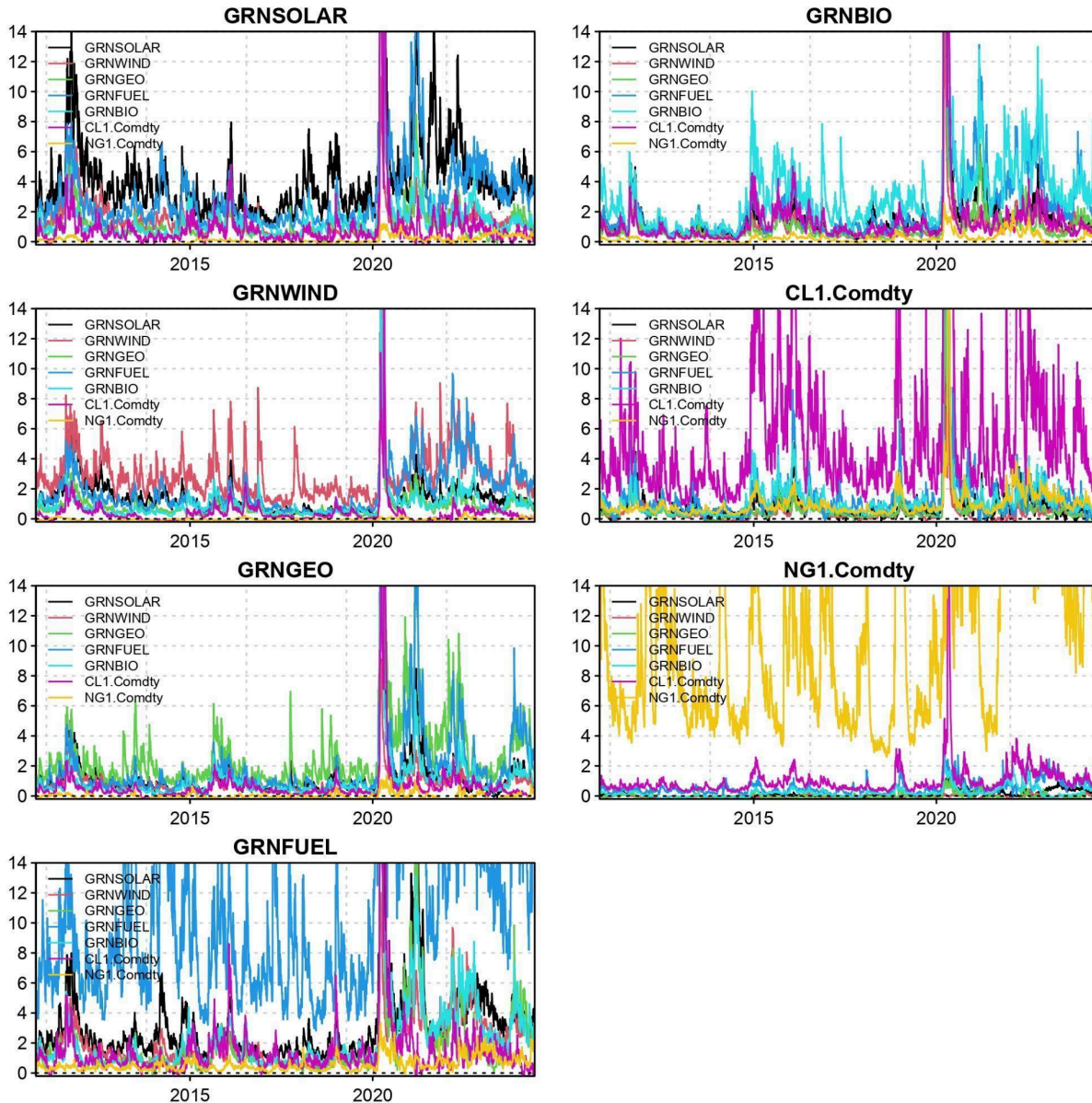
		SignBias	WARCH(20)	VaR	CVaR	VaR Dur.
Solar Energy		0.5337267	17.4601725	2.1569548	-740.1680	1.1169591
pvalues		0.5935652	0.0489603	0.1419257	0.9740	0.9151526
Wind Energy		1.3846951	4.9296188	1.7369845	-734.3662	1.1052632
pvalues		0.1662357	0.9586209	0.1875214	0.7210	0.3873517
Geothermal Energy		0.5270450	7.2302645	0.0938136	-672.1168	0.9824561
pvalues		0.5981964	0.8022367	0.7593838	0.7090	0.3727513
Fuel Cell		0.4121531	4.9314805	0.0073945	-687.1659	1.0116959
pvalues		0.6802530	0.9585479	0.9314732	0.9610	0.2046119
Bio Fuel		0.1283900	3.0109352	0.1021701	-696.1254	1.0292398
pvalues		0.8978478	0.9969242	0.7492409	0.5970	0.3324904
Crude Oil		0.8729327	0.1339828	0.0049683	-681.1639	1.0000000
pvalues		0.3827608	1.0000000	0.9438069	0.2240	0.6759649
Natural Gas		1.4670880	7.0823666	2.1569548	-740.1680	1.1169591
pvalues		0.1424437	0.8159626	0.1419257	0.9950	0.1983629

7.1.1. Dynamic conditional variance–covariances

Figure 2 plots the dynamic conditional variances and covariances estimated by the univariate GARCH models, providing insight into the time-varying nature of asset volatility and inter-dependencies. The most salient feature across all seven panels is the dramatic and contemporaneous spike in both variance (an asset's own volatility) and covariance (its co-movement with other assets) around the 2020 period. This visually confirms that volatility is highly clustered and that dependencies escalate significantly during market-wide crises, reinforcing the stylized facts observed in the raw return data.

The plots clearly illustrate the distinct volatility regimes between conventional and clean energy. The conditional variance for Crude Oil and, to a lesser extent, Natural Gas is shown to be on a much higher scale than that of the five clean energy indices, with the 2020 shock amplifying this difference massively. Within the clean energy complex (e.g., in the Solar Energy or Wind Energy panels), the variances and covariances are more tightly clustered, though Fuel Cell consistently displays a higher conditional variance than its green counterparts. The clear time-varying nature of these relationships underscores the necessity of employing a dynamic framework to capture the evolving risk and correlation structure of the energy market.

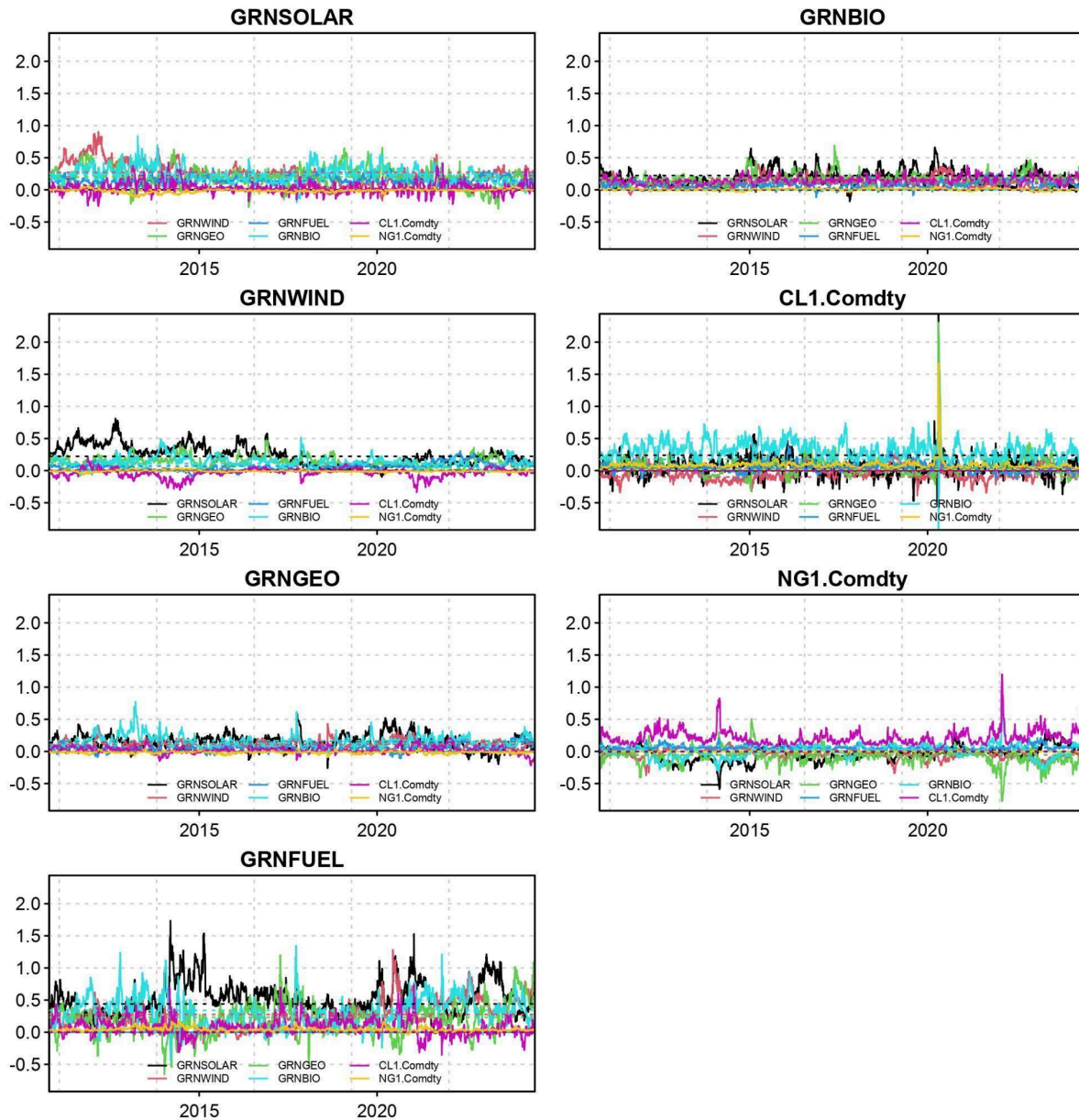
Figure 2: Dynamic conditional variance-covariances



7.1.2. Dynamic conditional betas

Figure 3 illustrates the dynamic conditional betas, highlighting the time-varying systematic risk of each asset relative to the others. The most prominent feature is their significant instability; the betas fluctuate constantly and exhibit extreme spikes during the 2020 crisis period. This high degree of time-variation strongly justifies the use of a dynamic conditional framework (Engle, 2002). The plots also reveal a critical divergence: betas relative to Crude Oil are generally positive but volatile, confirming it as a systematic risk factor. Conversely, betas relative to Natural Gas are exceptionally stable and hover near zero, indicating it is largely disconnected from the other energy assets and poses negligible systematic risk, suggesting it may be a valuable diversification tool (Sadorsky, 2012).

Figure 3: Dynamic conditional betas

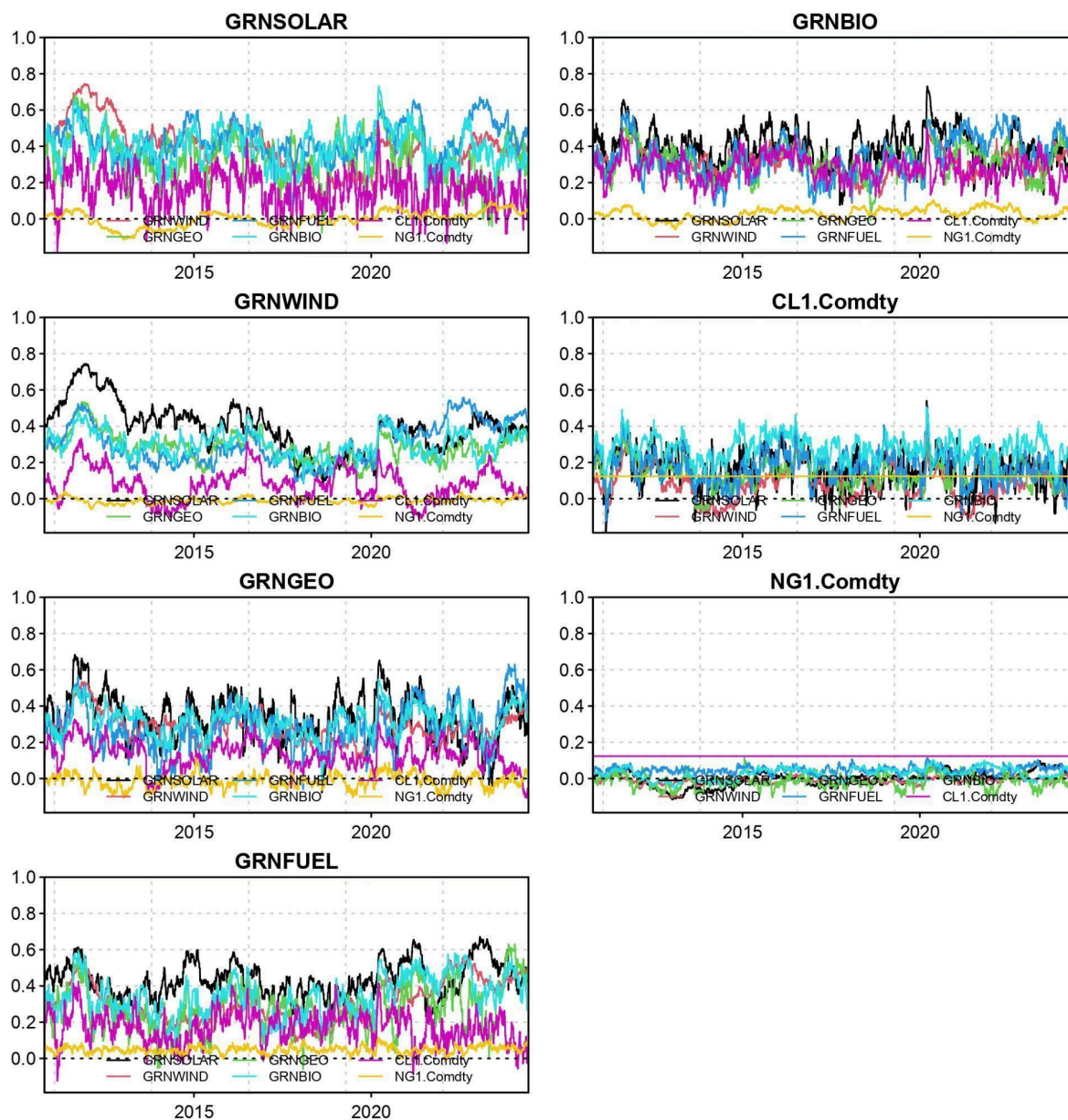


7.1.3. Dynamic conditional correlations

Figure 4 plots the dynamic conditional correlations (DCC) between all pairs of energy assets, providing a view of their co-movement with time, as captured by the DCC-GARCH model (Engle, 2002). The most evident finding is that all correlations are highly dynamic, reinforcing the unsuitability of static correlation measures. During the 2020 market crisis, there is a clear and sharp increase in correlations among almost all asset pairs, particularly within the clean energy complex (e.g., Solar Energy-Wind Energy, Solar Energy-Geothermal Energy) and between clean energy and crude oil. This illustrates a classic contagion effect, where assets co-move more strongly during periods of high systemic stress, diminishing diversification benefits.

The most striking and persistent finding, however, is the behavior of Natural Gas . The dynamic correlations between Natural Gas and all other assets (both clean and conventional) remain consistently near-zero throughout the entire sample period. Crucially, this near-zero correlation holds even during the 2020 crisis, a period when all other correlations spiked. This visually confirms that Natural Gas is almost completely decoupled from the rest of the energy market, operating independently and offering significant and stable diversification potential that is robust even during market-wide shocks.

Figure 4: Dynamic conditional correlations



7.1.4. Dynamic conditional R^2 decomposed measures

Figure 5: Dynamic conditional R^2 decomposed measures

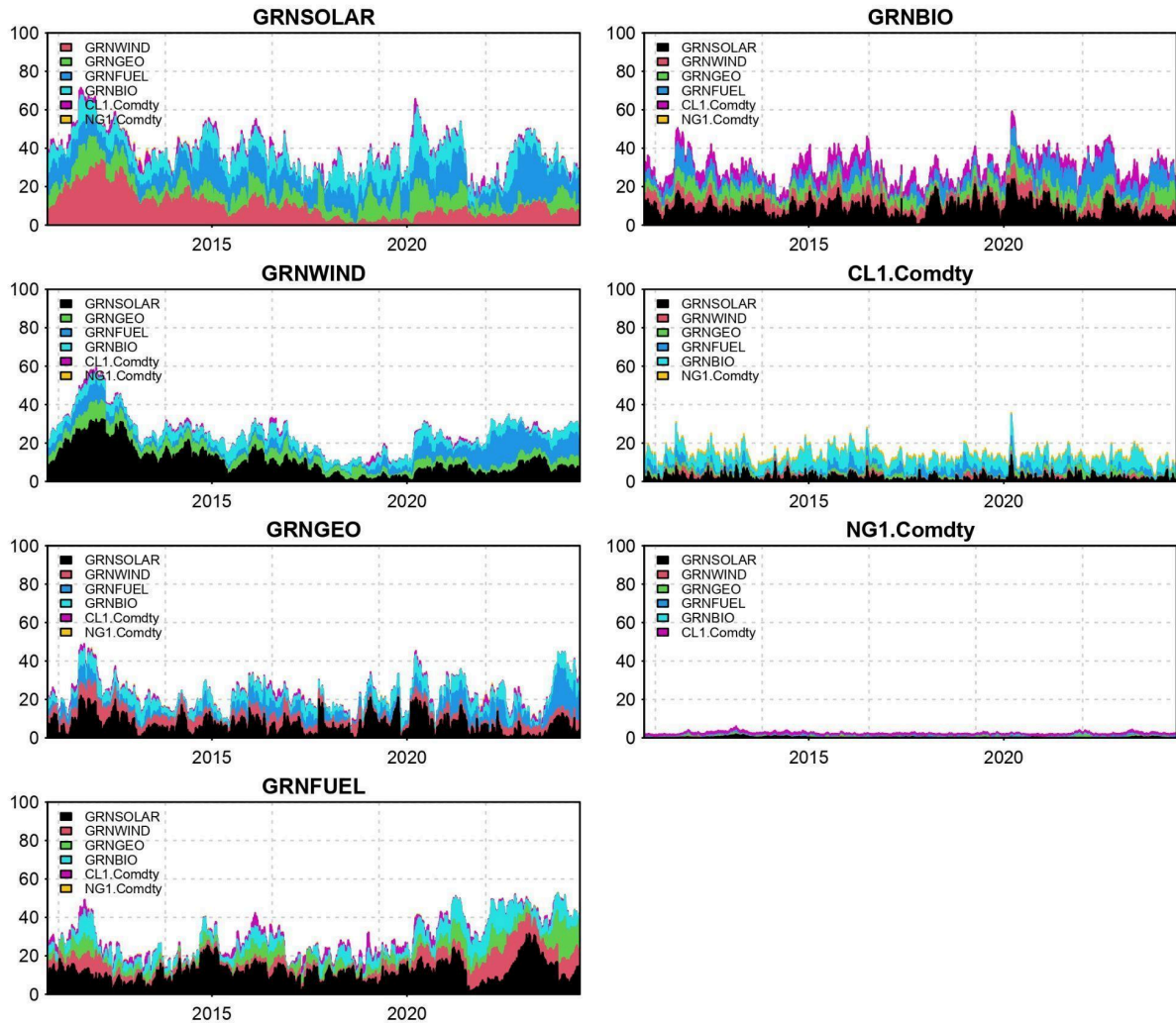


Figure 5 presents the total dynamic conditional R^2 decomposed connectedness measure for the entire system of seven energy assets. This plot is vital for our analysis, as it quantifies the overall system-wide risk and return propagation over time, calculated using the novel R^2 decomposed connectedness framework. The most striking feature is the measure's highly dynamic nature, demonstrating that system-wide connectedness is not constant but fluctuates significantly.

This dynamic behavior is most pronounced during the 2020 crisis, where the plot shows an unprecedented and dramatic surge. This spike indicates a rapid and massive increase in contagion, where the explanatory power of the assets for one another's returns peaked. In this period, virtually all assets began to move in lockstep, reflecting a system-wide shock that saturated the market. Following this 2020 peak, the connectedness level appears to moderate but remains structurally elevated compared to the pre-crisis period, suggesting a new, more interconnected regime in the energy market that has persisted, justifying the use of dynamic models (Engle, 2002).

7.2. R² decomposed connectedness approach

Table 3 : Averaged connectedness measures

	Solar Energy	Wind Energy	Geothermal Energy	Fuel Cell	Bio Fuel	Crude Oil	Natural Gas	FROM
Solar Energy	100.00	9.69	6.28	11.12	7.94	1.56	0.18	36.76
Wind Energy	10.20	100.00	4.00	5.17	4.37	0.68	0.05	24.47
Geothermal Energy	6.61	4.04	100.00	4.43	5.15	1.06	0.18	21.46
Fuel Cell	11.48	5.01	4.31	100.00	5.96	1.74	0.22	28.73
Bio Fuel	8.11	4.25	5.02	5.97	100.00	5.28	0.12	28.75
Crude Oil	1.71	0.73	1.11	1.84	5.69	100.00	1.41	12.50
Natural Gas	0.23	0.07	0.20	0.26	0.15	1.49	100.00	2.40
TO	38.34	23.79	20.91	28.79	29.26	11.81	2.16	155.07
Inc.Own	138.34	123.79	120.91	128.79	129.26	111.81	102.16	cTCI/TCI
NET	1.58	-0.67	-0.55	0.07	0.51	-0.69	-0.24	25.85/22.15
NPT	6.00	3.00	2.00	5.00	4.00	1.00	0.00	

Table 3 presents the static, or averaged, R² decomposed connectedness measures, which summarize the mean-spillover dynamics over the entire sample period. The total connectedness index (TCI) is found to be 22.15%, suggesting that, on average, a moderate portion of the assets' return variance can be attributed to spillovers from other assets in the system. This framework, based on the R² decomposition, allows for a precise attribution of these spillovers (Cocca et al., 2024). The analysis reveals a highly interconnected clean energy complex, with the five clean indices showing significantly higher 'TO' (spillovers received) and 'FROM' (spillovers transmitted) values than the conventional energy assets. Solar Energy (Solar) stands out as the most central asset, acting as both the largest recipient of spillovers (TO: 38.34) and the largest transmitter (FROM: 36.76), positioning it as the primary hub for risk propagation within the system.

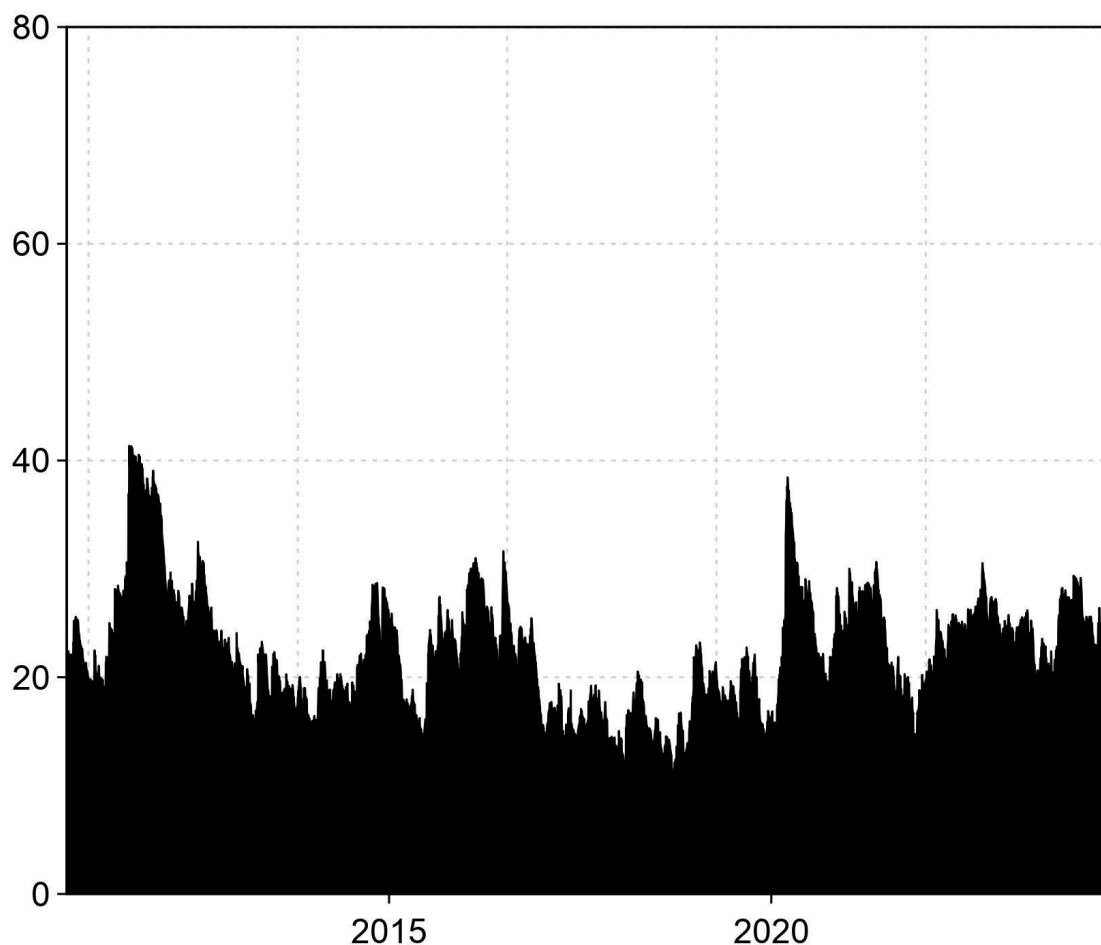
Further analysis of the net spillovers (the 'NET' row) helps identify which assets are, on average, net receivers or transmitters of risk (Diebold & Yilmaz, 2012). The system is clearly divided: Solar Energy (1.58), Bio Fuel (0.51), and Fuel Cell (0.07) are net receivers of shocks. In contrast, Wind Energy (-0.67), Geothermal Energy (-0.55), and both conventional energy assets—Crude Oil (-0.69) and Natural Gas (-0.24)—are net transmitters of risk. However, the role of conventional energy in this spillover system is minimal. While Crude Oil is a net transmitter, its overall interaction is low. Most notably, Natural Gas is almost entirely decoupled, with negligible 'TO' (2.16) and 'FROM' (2.40) values, confirming its strong diversification properties in an average sense.

7.2.1. Dynamic total connectedness

Figure 6 provides a dynamic view of each asset's role as a net transmitter or receiver of risk in the energy system. The plots illustrate the net spillovers for each of the seven assets over time; values above zero indicate the asset is a net transmitter of shocks, while values below zero indicate it is a net receiver. This dynamic approach, an evolution of static net spillover measures (Diebold & Yilmaz, 2012), reveals that an asset's role is not fixed and can change significantly, especially during crisis periods.

The plots show that most clean energy indices fluctuate around zero, alternating between being net transmitters and net receivers, though Solar Energy (Solar) spends significant time as a net receiver. A key finding is the behavior of conventional energy: Crude Oil is a persistent, though modest, net transmitter of risk to the system. The most striking result is for Natural Gas, which has a net connectedness plot that is almost flat and fixed at zero for the entire sample. This confirms that natural gas is completely decoupled from the system, neither transmitting nor receiving significant shocks, even during the 2020 crisis. This reinforces its robust diversification properties, as its role is stable and independent (Antonakakis et al., 2020).

Figure 6: Dynamic total connectedness



7.2.1. Net total directional connectedness

Figure 7: Net total directional connectedness

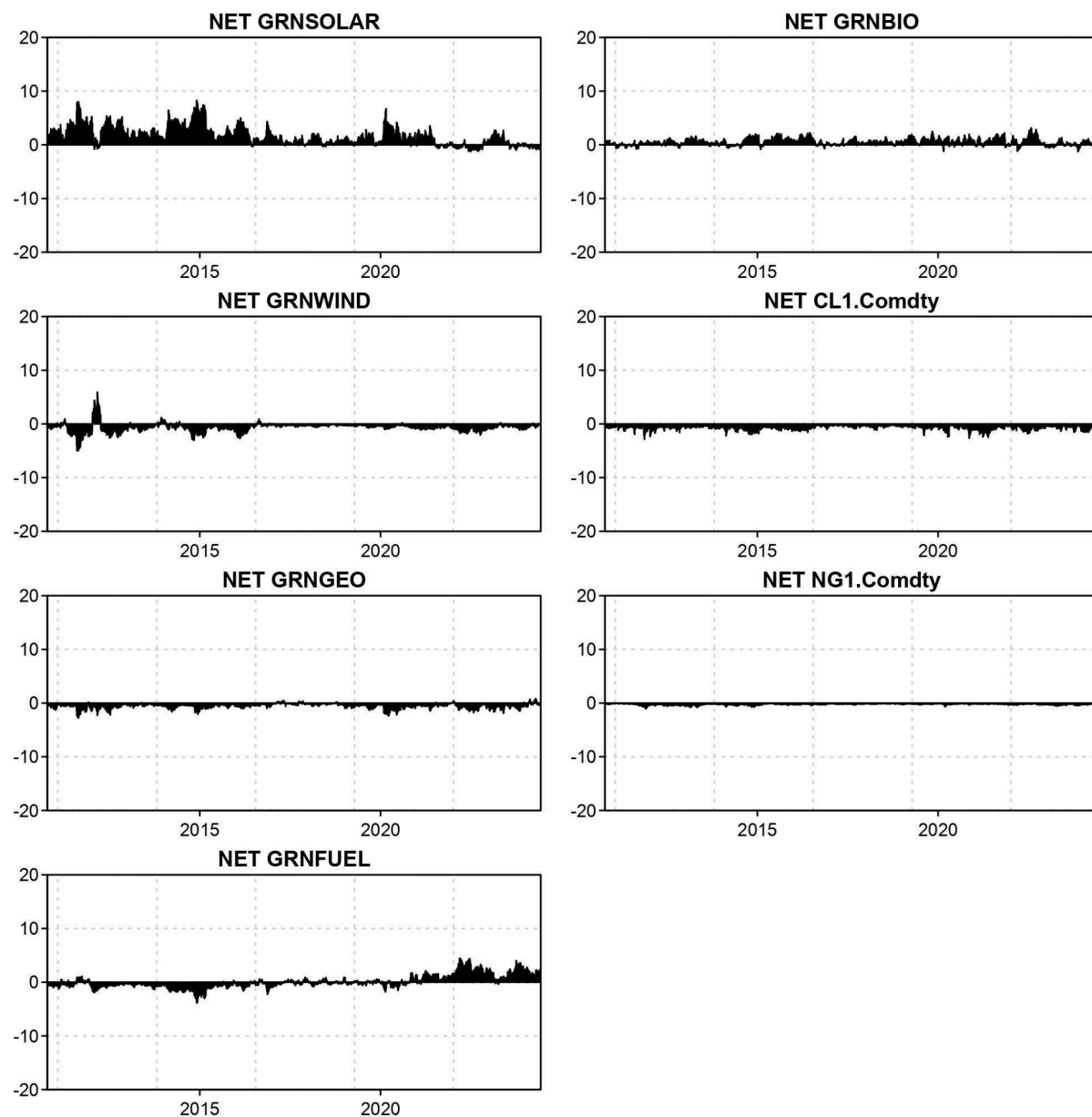


Figure 7 presents the dynamic net total directional connectedness for each of the seven energy assets. These plots are crucial as they illustrate the time-varying role of each asset as either a net transmitter of risk (a value above zero) or a net receiver of risk (a value below zero) to the entire system. This dynamic perspective, which advances on static measures, shows that an asset's role is not fixed and can shift significantly, particularly in response to market crises. The plots for the clean energy indices (Solar Energy, Wind Energy, etc.) largely fluctuate around the zero line, indicating they alternate between being net transmitters and receivers. Crude Oil is shown to be a persistent, albeit modest, net transmitter of risk. The most definitive finding comes from the plot for Natural

Gas , which is almost perfectly flat and locked at zero for the entire sample period. This visually confirms that natural gas is completely decoupled from the system, neither transmitting nor receiving shocks, and highlights its stable and robust diversification properties even during the 2020 crisis.

7.3. Portfolio and Risk Management

7.3.1. Hedging portfolios

Table 4: Hedge ratios

	Mean	Std.Dev.	5%	95%	HE	p-value	Return	Std.Dev	SR
Solar Energy/Wind Energy	0.466	0.176	0.215	0.823	0.192	0.997	0.085	0.296	0.288
Solar Energy/Geothermal Energy	0.456	0.210	0.174	0.840	0.171	0.938	0.122	0.300	0.406
Solar Energy/Fuel Cell	0.271	0.094	0.137	0.439	0.227	0.875	0.096	0.290	0.330
Solar Energy/Bio Fuel	0.504	0.229	0.226	0.972	0.220	0.830	0.113	0.291	0.387
Solar Energy/Crude Oil	0.156	0.110	0.003	0.370	0.043	0.851	0.120	0.322	0.372
Solar Energy/Natural Gas	0.000	0.030	-0.055	0.044	0.002	0.000	0.120	0.329	0.364
Wind Energy/Solar Energy	0.351	0.152	0.122	0.604	0.198	0.953	0.028	0.243	0.116
Wind Energy/Geothermal Energy	0.334	0.147	0.158	0.626	0.100	0.938	0.061	0.257	0.237
Wind Energy/Fuel Cell	0.164	0.079	0.065	0.318	0.147	0.875	0.066	0.250	0.262
Wind Energy/Bio Fuel	0.337	0.168	0.148	0.699	0.117	0.830	0.058	0.254	0.229
Wind Energy/Crude Oil	0.060	0.081	-0.069	0.220	-0.002	0.851	0.070	0.271	0.260
Wind Energy/Natural Gas	-0.005	0.008	-0.019	0.008	0.000	0.000	0.066	0.271	0.245
Geothermal Energy/Solar Energy	0.270	0.128	0.091	0.509	0.172	0.953	-0.046	0.243	-0.189
Geothermal Energy/Wind Energy	0.264	0.107	0.132	0.485	0.104	0.997	-0.013	0.252	-0.050
Geothermal Energy/Fuel Cell	0.138	0.076	0.031	0.276	0.135	0.875	-0.002	0.248	-0.007
Geothermal Energy/Bio Fuel	0.311	0.139	0.124	0.572	0.149	0.830	0.012	0.246	0.047
Geothermal Energy/Crude Oil	0.096	0.070	-0.005	0.222	-0.075	0.851	0.022	0.277	0.080
Geothermal Energy/Natural Gas	-0.006	0.021	-0.039	0.028	0.003	0.000	0.017	0.266	0.065
Fuel Cell/Solar Energy	0.756	0.289	0.388	1.282	0.224	0.953	-0.099	0.483	-0.206
Fuel Cell/Wind Energy	0.627	0.308	0.245	1.211	0.133	0.997	-0.035	0.511	-0.068
Fuel Cell/Geothermal Energy	0.617	0.294	0.170	1.145	0.116	0.938	0.013	0.516	0.024
Fuel Cell/Bio Fuel	0.715	0.308	0.278	1.261	0.163	0.830	0.003	0.502	0.007

Fuel Cell/Crude Oil	0.268	0.168	0.048	0.552	-0.002	0.851	0.018	0.549	0.032
Fuel Cell/Natural Gas	0.052	0.030	0.015	0.101	0.005	0.000	-0.003	0.547	-0.006
Bio Fuel/Solar Energy	0.351	0.165	0.133	0.670	0.257	0.953	-0.084	0.253	-0.331
Bio Fuel/Wind Energy	0.312	0.134	0.129	0.542	0.144	0.997	-0.062	0.272	-0.227
Bio Fuel/Geothermal Energy	0.359	0.152	0.155	0.625	0.148	0.938	-0.033	0.271	-0.122
Bio Fuel/Fuel Cell	0.185	0.088	0.054	0.335	0.197	0.875	-0.049	0.263	-0.188
Bio Fuel/Crude Oil	0.218	0.079	0.097	0.359	-0.094	0.851	-0.005	0.307	-0.017
Bio Fuel/Natural Gas	0.019	0.022	-0.010	0.055	0.007	0.000	-0.035	0.292	-0.118
Crude Oil/Solar Energy	0.204	0.214	0.002	0.522	0.038	0.953	-0.053	0.970	-0.055
Crude Oil/Wind Energy	0.131	0.215	-0.061	0.395	0.007	0.997	0.018	0.986	0.019
Crude Oil/Geothermal Energy	0.221	0.227	-0.005	0.470	0.022	0.938	-0.022	0.978	-0.022
Crude Oil/Fuel Cell	0.135	0.136	0.013	0.323	0.026	0.875	-0.028	0.976	-0.029
Crude Oil/Bio Fuel	0.389	0.179	0.174	0.657	0.011	0.830	0.009	0.984	0.009
Crude Oil/Natural Gas	0.094	0.085	0.041	0.159	0.006	0.000	-0.002	0.986	-0.002
Natural Gas/Solar Energy	0.003	0.078	-0.127	0.134	0.001	0.953	0.028	0.572	0.049
Natural Gas/Wind Energy	-0.018	0.038	-0.082	0.037	0.000	0.997	0.031	0.573	0.054
Natural Gas/Geothermal Energy	-0.033	0.098	-0.201	0.111	0.003	0.938	0.035	0.572	0.062
Natural Gas/Fuel Cell	0.052	0.030	0.013	0.111	0.004	0.875	0.027	0.572	0.048
Natural Gas/Bio Fuel	0.051	0.078	-0.075	0.174	0.001	0.830	0.028	0.572	0.049
Natural Gas/Crude Oil	0.205	0.100	0.096	0.368	-0.005	0.851	0.031	0.574	0.053

Table 4 presents the summary statistics for the dynamic hedge ratios, which are essential for optimal risk management between pairs of assets. The 'Mean' column shows the average optimal hedge ratio; for instance, the 0.466 value for Solar Energy/Wind Energy suggests that to hedge a \$1 long position in Solar Energy, an investor should short \$0.466 of Wind Energy. The low mean hedge ratio for Solar Energy/Natural Gas (0.000) and Crude Oil/Natural Gas (0.094) is striking, indicating that very little Natural Gas is required to hedge positions in Solar or Crude Oil, which confirms its decoupling.

The table also evaluates the performance of these hedging strategies. The 'Std.Dev.' column shows the high volatility of the hedge ratios (e.g., 0.176 for Solar Energy/Wind Energy), reinforcing the need for dynamic models. The 'HE' (Hedging Effectiveness) column is critical, showing the percentage reduction in portfolio variance. Hedging effectiveness is generally low to moderate across most pairs. Notably, the p-values ('p-value') for the hedging effectiveness are overwhelmingly high (e.g., 0.997 for Solar Energy/Wind Energy, 0.938 for Solar Energy/Geothermal Energy), indicating that the risk reduction is statistically insignificant for most pairs. The only exception is the hedge between Solar Energy and Natural Gas, which has a p-value of 0.000, suggesting its minimal hedging effect is, ironically, the most statistically reliable. Finally, the Sharpe Ratios ('SR')

for the hedged portfolios are modest, with the Solar Energy/Geothermal Energy pair showing a relatively attractive risk-adjusted return (0.406).

Table 5: Multivariate hedging portfolios

	Mean	Std.Dev.	5%	95%	HE	p-value	Return	Risk	SR
Solar Energy/Wind Energy	0.254	0.159	0.039	0.550	0.348	0.000	0.077	0.266	0.290
Solar Energy/Geothermal Energy	0.182	0.147	-0.056	0.444	0.348	0.000	0.077	0.266	0.290
Solar Energy/Fuel Cell	0.156	0.062	0.060	0.261	0.348	0.000	0.077	0.266	0.290
Solar Energy/Bio Fuel	0.225	0.126	0.046	0.450	0.348	0.000	0.077	0.266	0.290
Solar Energy/Crude Oil	0.035	0.086	-0.095	0.191	0.348	0.000	0.077	0.266	0.290
Solar Energy/Natural Gas	-0.013	0.026	-0.064	0.027	0.348	0.000	0.077	0.266	0.290
Wind Energy/Solar Energy	0.234	0.152	0.028	0.496	0.261	0.000	0.051	0.233	0.219
Wind Energy/Geothermal Energy	0.135	0.079	0.013	0.273	0.261	0.000	0.051	0.233	0.219
Wind Energy/Fuel Cell	0.064	0.061	-0.010	0.190	0.261	0.000	0.051	0.233	0.219
Wind Energy/Bio Fuel	0.128	0.080	0.013	0.274	0.261	0.000	0.051	0.233	0.219
Wind Energy/Crude Oil	-0.034	0.070	-0.175	0.061	0.261	0.000	0.051	0.233	0.219
Wind Energy/Natural Gas	-0.006	0.014	-0.026	0.023	0.261	0.000	0.051	0.233	0.219
Geothermal Energy/Solar Energy	0.139	0.114	-0.042	0.345	0.205	0.000	-0.020	0.238	-0.083
Geothermal Energy/Wind Energy	0.112	0.064	0.012	0.228	0.205	0.000	-0.020	0.238	-0.083
Geothermal Energy/Fuel Cell	0.054	0.057	-0.032	0.161	0.205	0.000	-0.020	0.238	-0.083
Geothermal Energy/Bio Fuel	0.153	0.092	0.046	0.323	0.205	0.000	-0.020	0.238	-0.083
Geothermal Energy/Crude Oil	0.024	0.052	-0.058	0.108	0.205	0.000	-0.020	0.238	-0.083
Geothermal Energy/Natural Gas	-0.013	0.019	-0.044	0.014	0.205	0.000	-0.020	0.238	-0.083
Fuel Cell/Solar Energy	0.512	0.258	0.179	1.016	0.303	0.000	-0.079	0.458	-0.172
Fuel Cell/Wind Energy	0.239	0.218	-0.034	0.648	0.303	0.000	-0.079	0.458	-0.172
Fuel Cell/Geothermal Energy	0.207	0.238	-0.172	0.616	0.303	0.000	-0.079	0.458	-0.172
Fuel Cell/Bio Fuel	0.298	0.224	-0.004	0.696	0.303	0.000	-0.079	0.458	-0.172
Fuel Cell/Crude Oil	0.088	0.130	-0.108	0.312	0.303	0.000	-0.079	0.458	-0.172
Fuel Cell/Natural Gas	0.042	0.030	0.004	0.098	0.303	0.000	-0.079	0.458	-0.172
Bio Fuel/Solar Energy	0.188	0.123	0.031	0.419	0.269	0.000	-0.061	0.251	-0.245
Bio Fuel/Wind Energy	0.120	0.080	0.009	0.272	0.269	0.000	-0.061	0.251	-0.245
Bio Fuel/Geothermal Energy	0.160	0.088	0.052	0.330	0.269	0.000	-0.061	0.251	-0.245

Bio Fuel/Fuel Cell	0.076	0.057	-0.001	0.183	0.269	0.000	-0.061	0.251	-0.245
Bio Fuel/Crude Oil	0.149	0.056	0.065	0.249	0.269	0.000	-0.061	0.251	-0.245
Bio Fuel/Natural Gas	0.002	0.014	-0.021	0.026	0.269	0.000	-0.061	0.251	-0.245
Crude Oil/Solar Energy	0.054	0.154	-0.165	0.259	0.032	0.338	-0.129	0.973	-0.133
Crude Oil/Wind Energy	-0.036	0.107	-0.184	0.159	0.032	0.338	-0.129	0.973	-0.133
Crude Oil/Geothermal Energy	0.065	0.156	-0.105	0.246	0.032	0.338	-0.129	0.973	-0.133
Crude Oil/Fuel Cell	0.053	0.076	-0.049	0.186	0.032	0.338	-0.129	0.973	-0.133
Crude Oil/Bio Fuel	0.318	0.128	0.143	0.545	0.032	0.338	-0.129	0.973	-0.133
Crude Oil/Natural Gas	0.085	0.073	0.037	0.136	0.032	0.338	-0.129	0.973	-0.133
Natural Gas/Solar Energy	-0.047	0.106	-0.223	0.134	0.005	0.889	0.057	0.571	0.100
Natural Gas/Wind Energy	-0.039	0.071	-0.167	0.062	0.005	0.889	0.057	0.571	0.100
Natural Gas/Geothermal Energy	-0.074	0.119	-0.268	0.100	0.005	0.889	0.057	0.571	0.100
Natural Gas/Fuel Cell	0.056	0.038	0.004	0.129	0.005	0.889	0.057	0.571	0.100
Natural Gas/Bio Fuel	-0.002	0.097	-0.169	0.130	0.005	0.889	0.057	0.571	0.100
Natural Gas/Crude Oil	0.214	0.110	0.091	0.391	0.005	0.889	0.057	0.571	0.100

Table 5 provides the summary statistics for the multivariate hedging portfolios, which extend the bivariate approach by considering the optimal hedge for one asset against all other assets in the system. The 'Mean' column shows the average hedge ratio required for each asset in the hedging portfolio. For example, to hedge a \$1 position in Solar Energy (Solar), an investor would need to short \$0.254 of Wind Energy (Wind), \$0.182 of Geothermal Energy (Geothermal), and so on. The consistently smallest (and in one case, negative) hedge ratio is for Natural Gas, confirming its minimal role in hedging the other assets.

The 'HE' (Hedging Effectiveness) column is uniform for all assets hedging Solar Energy (0.348), indicating that this multivariate portfolio reduces 34.8% of the variance of a long position in Solar. This effectiveness is highly statistically significant, as shown by the 'p-value' of 0.000 for all these portfolios. This contrasts sharply with the bivariate results (from Table 4), which were mostly insignificant. This demonstrates the clear superiority of a multivariate hedging approach, which provides substantial and statistically significant risk reduction. The 'Risk' (standard deviation) of the hedged Solar Energy portfolio is 0.266, and it yields a Sharpe Ratio ('SR') of 0.290. A notable finding is that hedging Crude Oil with the other assets yields very low hedging effectiveness (0.032), and the resulting portfolio has a negative Sharpe Ratio (-0.133), making it an unattractive strategy.

7.3.2. Minimum risk portfolios

Table 6: Optimal bivariate portfolio weights

	Mean	Std.Dev.	5%	95%	HE	p-value	SR
Solar Energy/Wind Energy	0.380	0.176	0.113	0.697	0.449	0.000	0.297
Solar Energy/Geothermal Energy	0.318	0.167	0.064	0.621	0.465	0.000	0.299
Solar Energy/Fuel Cell	0.840	0.155	0.548	1.000	0.078	0.017	0.379
Solar Energy/Bio Fuel	0.386	0.242	0.008	0.798	0.396	0.000	0.230
Solar Energy/Crude Oil	0.546	0.199	0.231	0.879	0.151	0.000	0.099
Solar Energy/Natural Gas	0.710	0.149	0.407	0.904	0.336	0.000	0.758
Wind Energy/Solar Energy	0.620	0.176	0.303	0.887	0.185	0.000	0.297
Wind Energy/Geothermal Energy	0.428	0.191	0.132	0.776	0.406	0.000	0.218
Wind Energy/Fuel Cell	0.861	0.118	0.645	1.000	0.057	0.084	0.137
Wind Energy/Bio Fuel	0.489	0.227	0.078	0.812	0.342	0.000	0.167
Wind Energy/Crude Oil	0.620	0.170	0.323	0.869	0.295	0.000	0.255
Wind Energy/Natural Gas	0.764	0.120	0.544	0.927	0.236	0.000	0.488
Geothermal Energy/Solar Energy	0.682	0.167	0.379	0.936	0.184	0.000	0.299
Geothermal Energy/Wind Energy	0.572	0.191	0.224	0.868	0.387	0.000	0.218
Geothermal Energy/Fuel Cell	0.894	0.091	0.718	1.000	0.049	0.145	0.115
Geothermal Energy/Bio Fuel	0.546	0.211	0.202	0.888	0.289	0.000	0.018
Geothermal Energy/Crude Oil	0.677	0.175	0.335	0.925	0.079	0.016	-0.037
Geothermal Energy/Natural Gas	0.792	0.131	0.512	0.946	0.261	0.000	0.377
Fuel Cell/Solar Energy	0.160	0.155	0.000	0.452	0.668	0.000	0.379
Fuel Cell/Wind Energy	0.139	0.118	0.000	0.355	0.770	0.000	0.137
Fuel Cell/Geothermal Energy	0.106	0.091	0.000	0.282	0.775	0.000	0.115
Fuel Cell/Bio Fuel	0.129	0.130	0.000	0.388	0.731	0.000	-0.071
Fuel Cell/Crude Oil	0.301	0.186	0.058	0.659	0.444	0.000	-0.446
Fuel Cell/Natural Gas	0.497	0.171	0.203	0.765	0.511	0.000	0.311
Bio Fuel/Solar Energy	0.614	0.242	0.202	0.992	0.240	0.000	0.230
Bio Fuel/Wind Energy	0.511	0.227	0.188	0.922	0.440	0.000	0.167
Bio Fuel/Geothermal Energy	0.454	0.211	0.112	0.798	0.413	0.000	0.018
Bio Fuel/Fuel Cell	0.871	0.130	0.612	1.000	0.060	0.071	-0.071
Bio Fuel/Crude Oil	0.668	0.171	0.364	0.915	0.118	0.000	-0.157
Bio Fuel/Natural Gas	0.776	0.136	0.506	0.939	0.262	0.000	0.106
Crude Oil/Solar Energy	0.454	0.199	0.121	0.769	0.906	0.000	0.099

Solar Energy	0.079	0.044	0.000	0.146	0.159	0.000	0.013
Wind Energy	0.155	0.033	0.106	0.208	-0.243	0.000	0.013
Geothermal Energy	0.153	0.025	0.114	0.194	-0.282	0.000	0.013
Fuel Cell	0.099	0.039	0.021	0.159	0.697	0.000	0.013
Bio Fuel	0.078	0.031	0.027	0.126	-0.058	0.096	0.013
Crude Oil	0.174	0.028	0.132	0.233	0.907	0.000	0.013
Natural Gas	0.262	0.024	0.227	0.302	0.722	0.000	0.013
Minimum connectedness portfolio (MPP)							
Solar Energy	0.105	0.023	0.069	0.145	0.305	0.000	0.093
Wind Energy	0.143	0.015	0.120	0.164	-0.028	0.421	0.093
Geothermal Energy	0.147	0.011	0.128	0.167	-0.060	0.089	0.093
Fuel Cell	0.130	0.021	0.090	0.158	0.750	0.000	0.093
Bio Fuel	0.124	0.013	0.102	0.145	0.125	0.000	0.093
Crude Oil	0.164	0.010	0.149	0.183	0.923	0.000	0.093
Natural Gas	0.188	0.010	0.173	0.205	0.770	0.000	0.093
Minimum bivariate R ² portfolio (MRP)							
Solar Energy	0.081	0.041	0.000	0.141	0.166	0.000	-0.001
Wind Energy	0.145	0.026	0.110	0.182	-0.233	0.000	-0.001
Geothermal Energy	0.147	0.017	0.120	0.178	-0.271	0.000	-0.001
Fuel Cell	0.119	0.036	0.048	0.167	0.700	0.000	-0.001
Bio Fuel	0.106	0.023	0.065	0.140	-0.050	0.156	-0.001
Crude Oil	0.183	0.018	0.157	0.220	0.908	0.000	-0.001
Natural Gas	0.220	0.023	0.188	0.258	0.724	0.000	-0.001
Minimum R ² decomposed connectedness portfolio (M2P)							
Solar Energy	0.123	0.011	0.105	0.144	0.349	0.000	0.122
Wind Energy	0.141	0.009	0.126	0.155	0.038	0.255	0.122
Geothermal Energy	0.144	0.007	0.133	0.157	0.008	0.808	0.122
Fuel Cell	0.135	0.011	0.113	0.151	0.766	0.000	0.122
Bio Fuel	0.133	0.007	0.121	0.145	0.181	0.000	0.122
Crude Oil	0.155	0.006	0.146	0.167	0.928	0.000	0.122
Natural Gas	0.169	0.007	0.159	0.179	0.785	0.000	0.122

Table 7 provides a comprehensive multivariate portfolio analysis comparing five optimization strategies: Minimum Variance Portfolio (MVP), Minimum Correlation Portfolio (MCP), Minimum Connectedness Portfolio (MPP), Minimum Bivariate R² Portfolio (MRP), and Minimum R² Decomposed Connectedness Portfolio (M2P). Across all strategies, MVP portfolios display high Sharpe Ratios (SR = 0.368) and highly concentrated allocations (high HE), signifying strong risk-adjusted returns but potential lack of diversification. MCP and MPP spread investments more evenly, with lower SRs and Herfindahl values, suggesting increased diversification but slightly compromised performance.

MRP shows near-zero or negative SRs, indicating limited advantage in risk-adjusted terms. The newly added M2P portfolio in the second image demonstrates moderate mean allocations and consistent SRs (0.122) across all assets, signifying uniformly distributed risk-adjusted return profiles and moderate concentration (HE in the range 0.008–0.928). Statistically, most p-values are significant, supporting the robustness of portfolio weights. The full analysis indicates that MVP may be best for maximizing risk-adjusted returns, whereas MCP, MPP, MRP, and M2P strategies cater to investors focusing on diversification, lower systemic risk, and interconnectedness among assets, each carrying trade-offs between concentration and risk-adjusted return performance.

7.3.3. Portfolio performance

Table 8 : Portfolio performance

	MVP	MCP	MPP	M2P	MRP
Return	0.000737	0.000654	0.000703	0.000621	0.000725
StdDev	0.001672	0.003020	0.002746	0.003007	0.002656
Sharpe Ratio (StdDev)	0.440579	0.216500	0.255838	0.206555	0.272816
Sharpe Ratio (VaR)	0.279037	0.184189	0.210090	0.180332	0.218918
Sharpe Ratio (CVaR)	0.190078	0.110935	0.125341	0.106901	0.130944

Table 8 provides a comprehensive performance comparison of five different portfolio strategies: the Minimum Variance Portfolio (MVP), MCP, MPP, M2P, and the Minimum R² Portfolio (MRP). The traditional MVP is superior across all the metrics. The MVP, which is designed to minimize portfolio risk, has a standard deviation (StdDev) of 0.001672, much lower than any other strategy. In contrast, the other four portfolios display much greater risk, with standard deviations ranging from 0.002656 (MRP) to 0.003020 (MCP). Alongside the risk reduction, the MVP also has the highest mean return (0.000737).

Consequently, the MVP strategy is best in terms of risk-adjusted returns. Its standard Sharpe Ratio (0.440579) is much higher than the next best portfolio, the MRP (0.272816). This dominance is consistent across all risk-adjusted metrics, including the Sharpe Ratios based on VaR (0.279037) and CVaR (0.190078). These findings indicate that while other strategies like the MRP are being explored, the traditional Markowitz (1952) minimum variance approach provides the most risk-free and highest-performing portfolio in this analysis.

8. Drivers of the connectedness network.

8.1 Methodology and Data

This section examines what are the driving forces behind the total connectedness index of the 7 asset classes by using Time-Varying Granger Causality (TVGC) analysis. For this analysis we have chosen 5 driver variables which could Granger cause the total connectedness index. These 5 driver variables are US Economic Policy Uncertainty (EPU), Geopolitical Risk (GPR), Implied Volatility of S&P 500 (VIX), Carbon Emission Allowance Price (Carbon Price), Oil Volatility Index (OVX).

The 2 drivers which were chosen by us were Carbon Emission Allowance Price (Carbon Price), Oil Volatility Index (OVX) as the assets used to construct the TCI are energy related. This analysis is done using forward, rolling and recursive window estimation with a window size of 504. The analysis is done in STATA. The analysis is based on daily data of TCI and the 5 driver variables. The TCI is calculated for 7 assets which are Solar Energy, Wind Energy, Geothermal Energy, Fuel Cell, Bio Fuel, Crude Oil, Natural Gas. For this analysis the data we took was the daily last price from Bloomberg for all the seven assets and the driver variables except for the US Economic Policy Uncertainty (EPU) index which was taken from the official website:

<https://www.policyuncertainty.com>.

To examine how each of the driver variables can predict changes in the Total Connectedness Index (TCI), we have applied the Time-Varying Granger Causality (TVGC) approach of Shi, Su, and Phillips (2020). We have estimated the Wald statistics on different segments of the sample (Forward, Rolling, Recursive) which are then compared against bootstrapped critical values to determine the periods during which the driver variables significantly Granger-cause the TCI.

8.2 Results

Table 8.1: Wald Tests of Granger Causality

Causality	Forward			Rolling			Recursive		
	Wald	90th	95th	Wald	90th	95th	Wald	90th	95th
EPU	12.821	16.353	20.575	16.563	17.12	20.938	21.36	18.811	22.66
VIX	239.94 5	48.251	53.779	83.787	47.466	51.369	239.94 5	51.386	56.441
GPR	9.874	6.217	8.13	10.216	6.735	8.302	12.132	7.506	8.912
Carbon Price	15.336	7.056	9.379	16.575	7.405	9.365	16.793	7.638	9.959
OVX	221.37 5	20.479	23.914	107.12 6	19.747	22.909	221.37 5	20.942	24.256

1) Implied Volatility of S&P 500 (VIX) – Strongest Driver

VIX was found to be the strongest driver of the total connected index where all Wald statistics (Forward, Rolling, Recursive) substantially exceeded their 95th percentile critical values.

This means there is high significant causality, so equity market's uncertainty strongly predicts the connectedness dynamics of the 7 assets

2) Crude Oil Volatility Index (OVX) - Second Strongest Driver

OVX was found to be Second Strongest Driver after VIX, where all Wald statistics (Forward, Rolling, Recursive) exceeded their 95th percentile critical values.

This means Oil market Volatility is a significant predictor of network connectedness.

3) Carbon Price - Moderate Significance

Wald statistics moderately exceeded the 90th percentile and 95th percentile.

This means that Carbon price is a predictor of network connectedness, but not as big of a predictor as Implied Volatility of S&P 500 and Crude Oil Volatility Index.

4) Geopolitical Risk - Moderate Significance

Wald statistics for all moderately exceeded the 90th percentile and 95th percentile.

So Geopolitical Risk affects the connectedness mostly during geopolitical shocks periods.

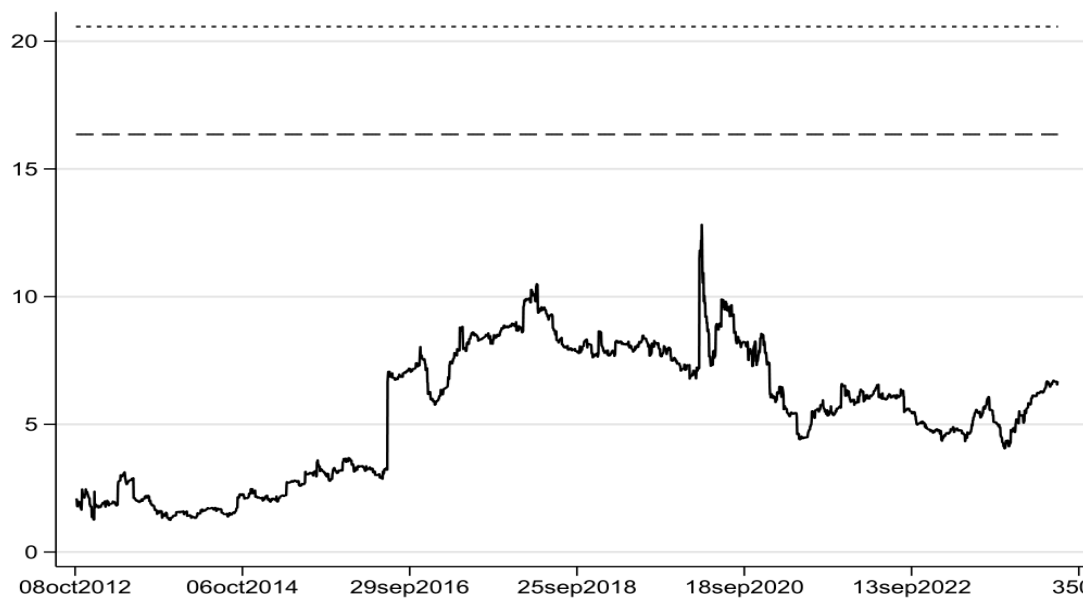
5) Economic Policy Uncertainty (EPU) – Weakest Driver

US Economic Policy Uncertainty was found to be the weakest driver for the TCI where only Wald statistics for Rolling window exceeded 90th percentile.

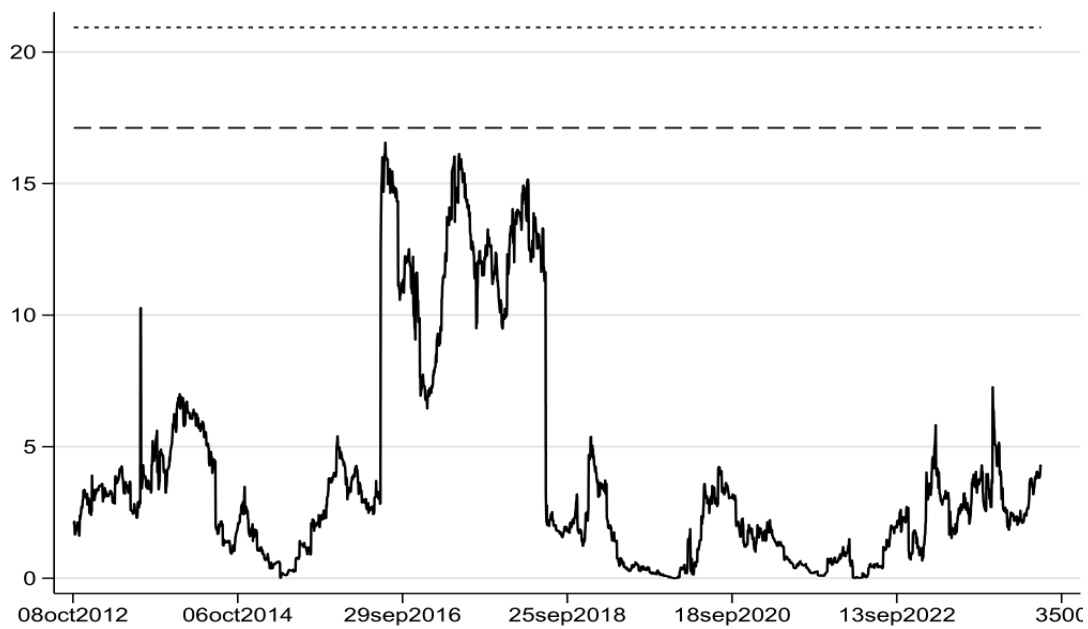
EPU does not influence the network connectedness

Time-Varying Dynamics

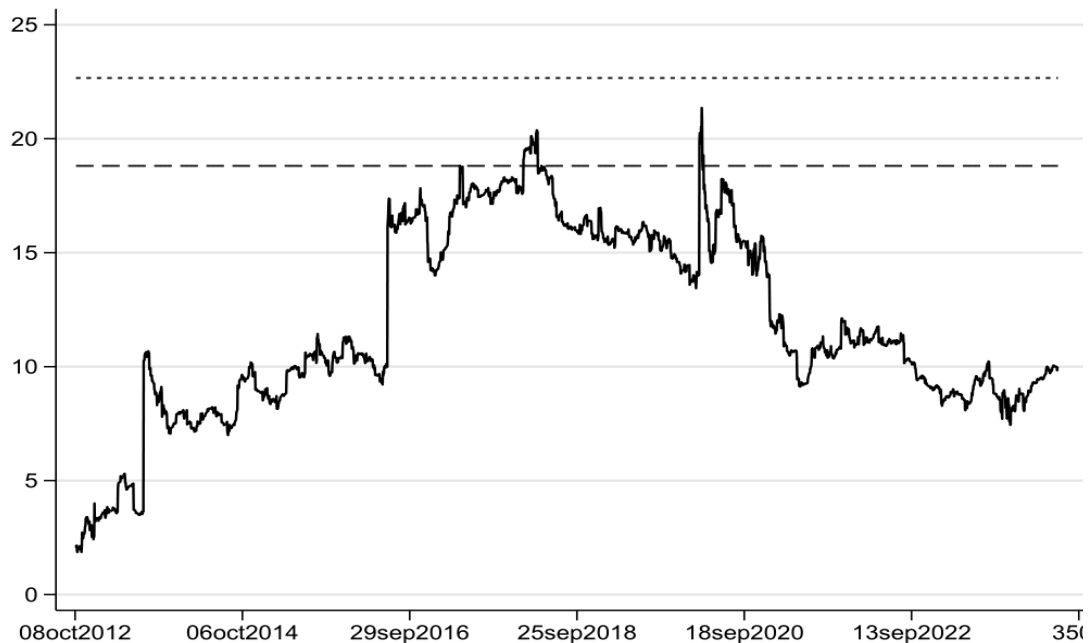
Figure 8.1: Time-Varying Granger Causality - EPU $\rightarrow$ TCI



Forward



Rolling



Recursive

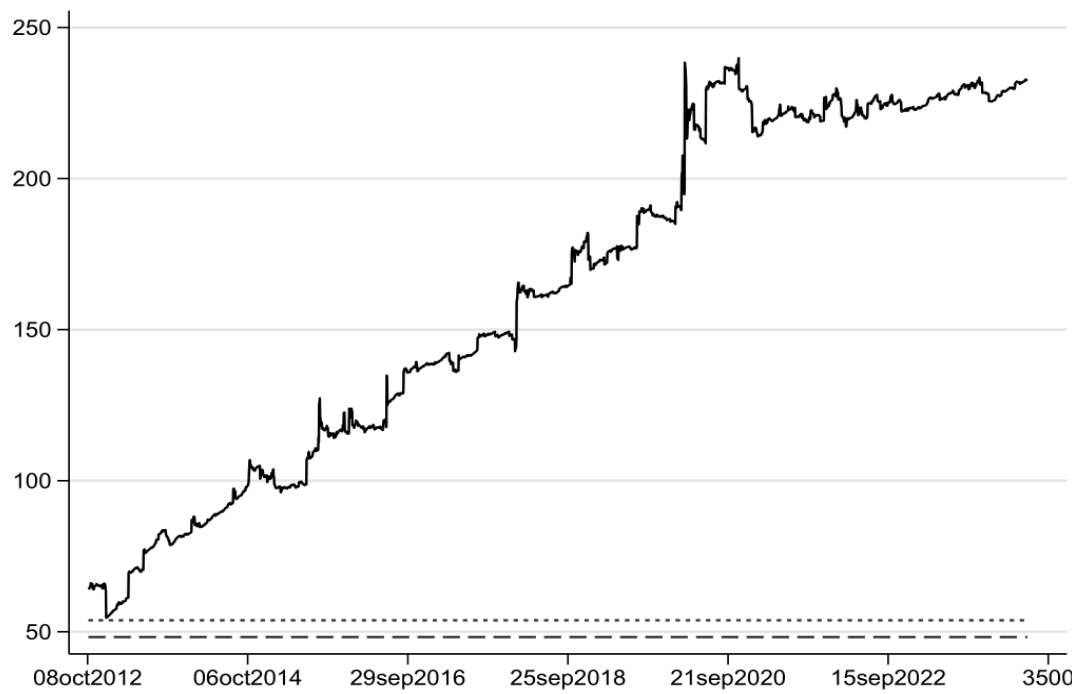
These graphs show time-varying Granger causality results for $EPU \rightarrow TCI$

These results show that Economic Policy Uncertainty has a consistently weak influence on the connectedness network.

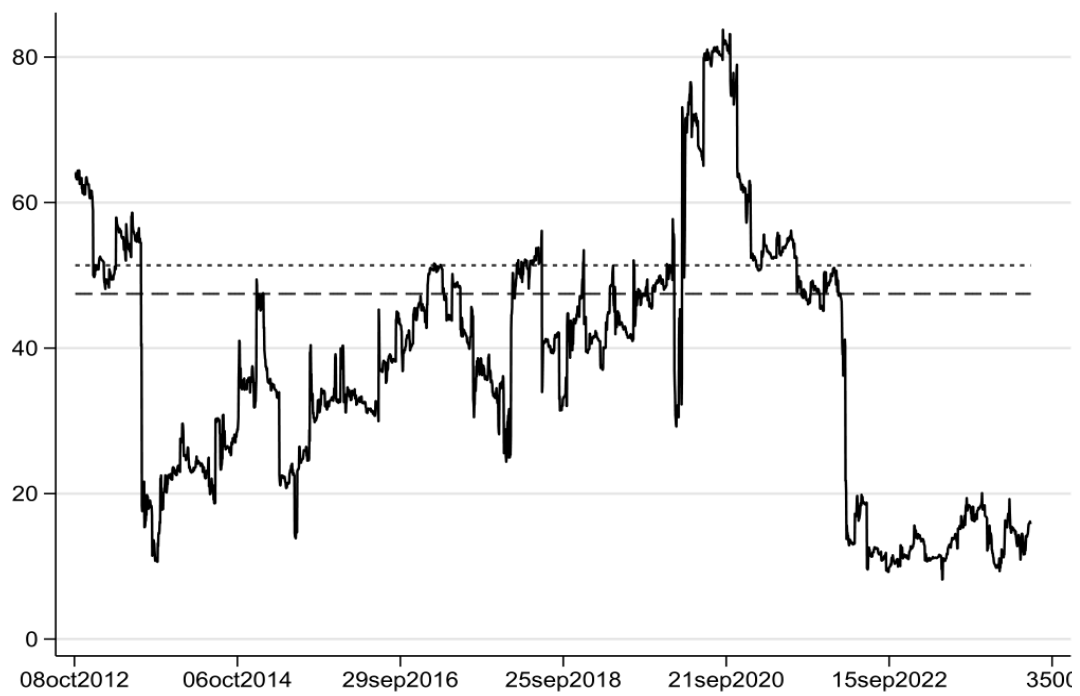
The forward and recursive windows show smooth and gradual movements but they never approach the critical lines, whereas the rolling window shows some short-term fluctuations especially around 2015 -2019 but even these spikes which crossed the 90th percentile critical value but none of these episodes sustained there long enough to indicate any meaningful causality.

Overall, here in all of these graphs we can see that EPU constantly stays below the two critical lines especially from 2012 to 2016 but during 2016 to late 2019 a strong, new relationship emerged but still that relationship vanished after 2019 and this concludes that EPU has weak influence on TCI.

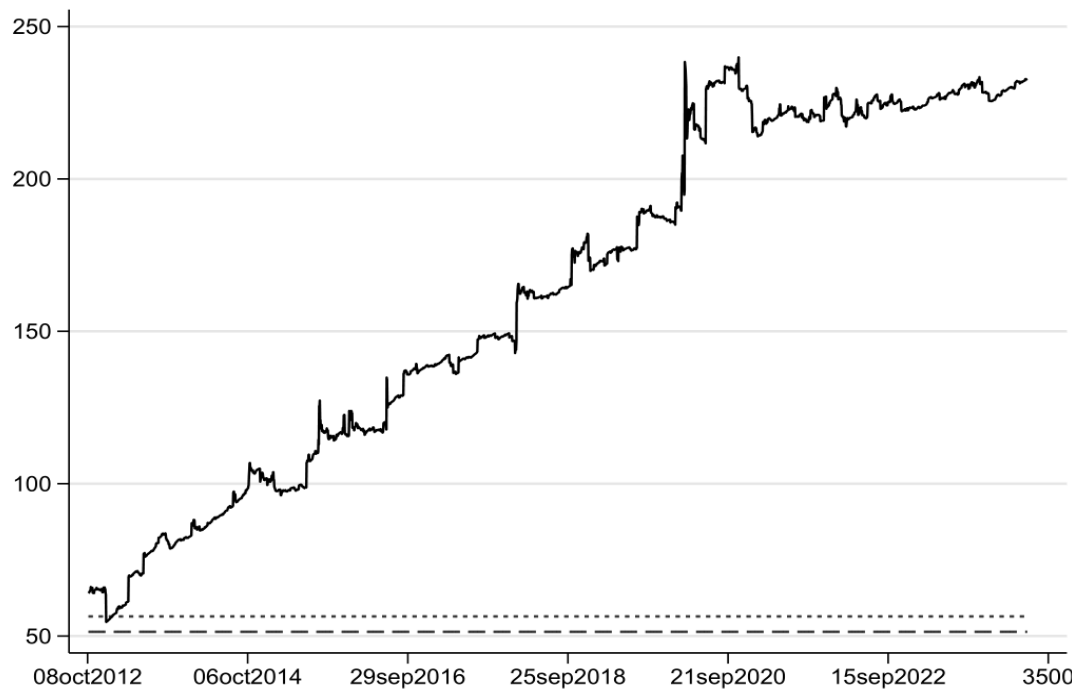
Figure 8.2: Time-Varying Granger Causality - VIX $\rightarrow$ TCI



Forward



Rolling



Recursive

These graphs show time-varying Granger causality results for $VIX \rightarrow TCI$

These results show that VIX strongly Granger Causes TCI across all window methods and it is the strongest and the most persistent driver of connectedness among all the 5 variables.

The Wald statistics remain above the 90% and 95% critical thresholds which indicates strong and statistically significant causality throughout the entire sample.

The Wald statistics keep on increasing from 2012 onwards and stay well above the critical values for the entire time. Also there was a sharp increase during the COVID-19 period showing the nature of VIX's influence on systemic spillovers. The rolling window showed more volatility, with recurrent spikes around 2015-2019 and a major jump during the pandemic.

Overall we can see that the graphs are well above the critical lines especially in the forward and recursive window and so VIX is a strong predictor of the total connectedness

Figure 8.3: Time-Varying Granger Causality - GPR $\rightarrow$ TCI



Forward



Rolling



Recursive

These graphs show time-varying Granger causality results for $GPR \rightarrow TCI$. In these graphs we can see that the GPR moderately stays above the critical values so they exert a moderate and recursive influence on the connectedness network.

In all estimation windows, the Wald statistics rise above the 90% critical threshold and in some periods even cross the 95% line particularly around major geopolitical events. This indicates that these peaks happen majorly where there is heightened global tensions and uncertainty. Since these time periods of peaks are not long lasting, causality quickly dampens after each spike and the causality remains well below the significance thresholds for long stretches, especially after 2021.

Overall, all three graphs indicate that the geopolitical risk does not function as a consistent or dominant driver of connectedness over the long term, it does have occasional but notable influence on TCI during times of geopolitical stress.

Figure 8.4: Time-Varying Granger Causality - Carbon Price → TCI



Forward



Rolling



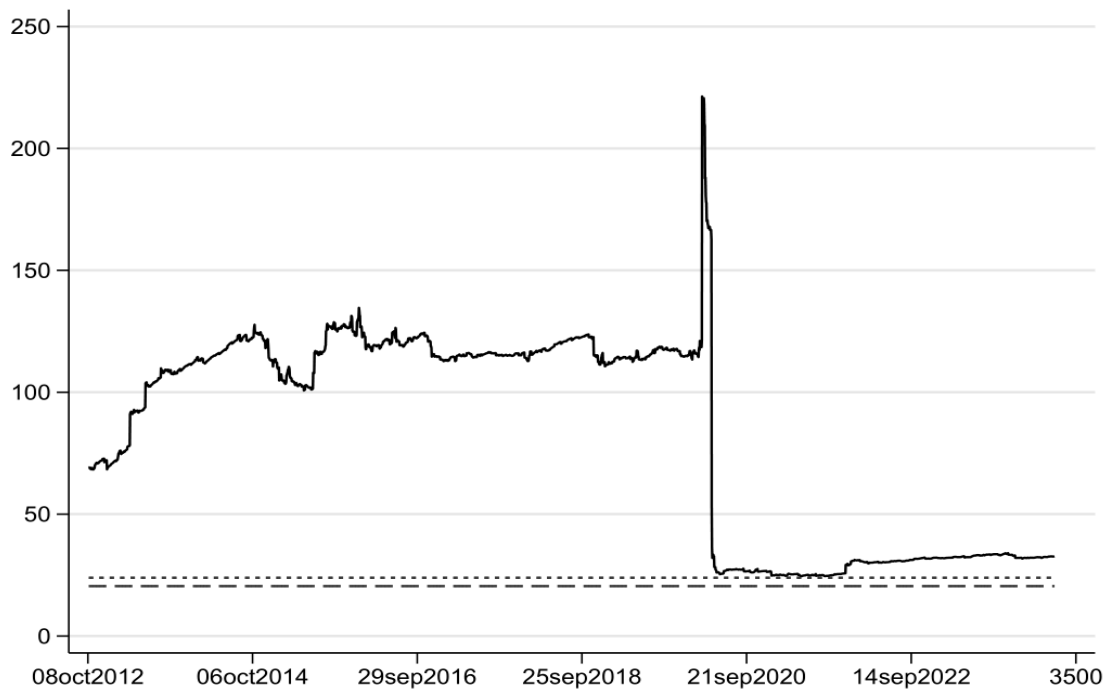
Recursive

These graphs show time-varying Granger causality results for Carbon Price → TCI. In these graphs we can see that the Carbon Price moderately stays above the critical values so this means that they exert a moderate but non-persistent influence on the total connectedness network.

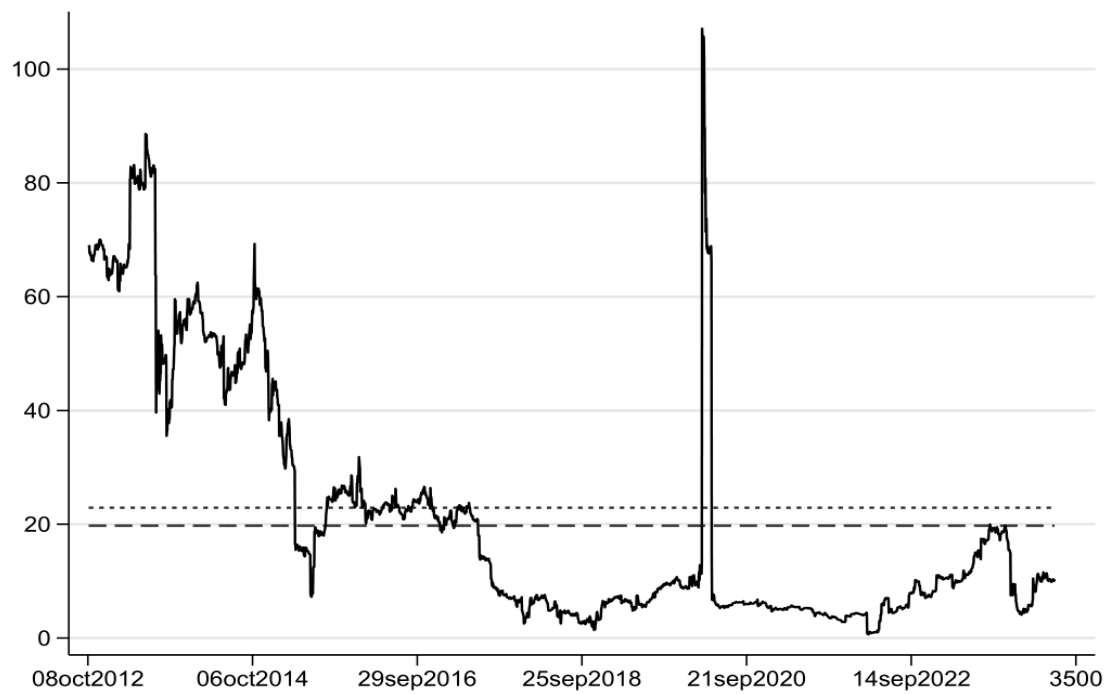
In all estimation windows, the Wald statistics occasionally rise above 90% critical values particularly between 2018 and early 2020 where it crosses 95% critical values but then quickly drops and does not stay above 90% critical values for long periods of time. These peaks occur when there are major fluctuations in carbon markets and a broader uncertainty on the environmental policy is present which suggests that carbon related shocks can temporarily strengthen cross asset spillovers. But after 2021 these causal episodes fade away quickly and Wald statistics remain well below both the critical values, which indicates a weakening relationship.

Overall, these graphs indicate that carbon prices can influence the Total Connectedness Index during times of significant market volatility but they are not a consistent or significant long-term driver of connectedness.

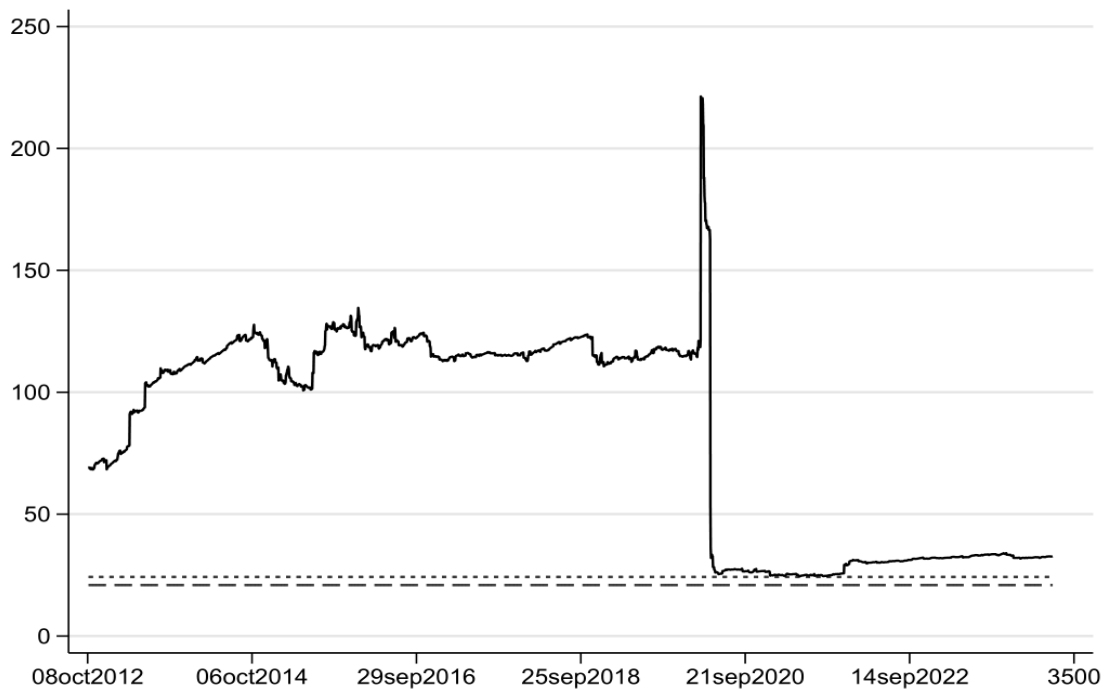
Figure 8.5: Time-Varying Granger Causality - OVX $\rightarrow$ TCI



Forward



Rolling



Recursive

These graphs show time-varying Granger causality results for $OVX \rightarrow TCI$. In these graphs we can clearly see that Oil market volatility is a strong and persistent driver of total connectedness network as the Wald statistics (in Forward and Recursive windows) stays above the critical values for the whole sample period.

A large spike occurs around 2020 which could be because of the oil market collapse because of COVID-19, which pushes the Wald statistic to extremely high levels, which shows how vulnerable the connectedness networks are to significant shocks to oil volatility. Although the Wald statistics drop sharply after the spike, it still remained close or above the critical thresholds for remaining time of the sample. The rolling window shows more short term fluctuations but still maintains significant causality for much of the sample period where in several episodes exceeded the 90% and 95% critical lines.

Overall, these results confirm that OVX is one of the major drivers of the Total Connected Index of the 7 assets particularly during times of intense market stress.

8. Conclusion and Policy Recommendations

This study explores the dynamic interconnectedness among clean energy indices and traditional energy markets using a robust DCC-GARCH R^2 decomposed connectedness method spanning data from 2010 to 2023 (different periods used for TVGC analysis). Key findings demonstrate pronounced time variation in connectivity, with systemic shocks intensifying during global events such as the COVID-19 pandemic and other geopolitical conflicts, confirmed by connectedness metrics and conditional correlations. Notably, while the broader Green Economy Index frequently acts as a net shock transmitter, subcomponents like Solar, Wind, and Geothermal energy mostly absorb shocks, highlighting risk diversification opportunities within the sector. Portfolio analysis shows that dynamically adjusting hedge ratios based on these decomposed connectedness measures achieves superior risk mitigation and portfolio performance compared to traditional static diversification.

The Time Varying Granger Causality (TVGC) analysis provides clear evidence that cross asset spillovers in our energy focused TCI are primarily driven by market based volatility indicators. Both Implied Volatility of S&P 500 (VIX) and Crude Oil Volatility Index (OVX) show a strong and persistent influence and suggest that they have good predictive power of the TCI of the 7 assets. Carbon price and geopolitical risk moderately contribute, with huge spikes during periods of market stress or policy shifts. US economic policy uncertainty shows weak and short lived effects. These results basically say that the shocks from equity volatility (VIX) and oil market volatility (OVX) are transmitted most strongly through the system.

Given this evolving risk landscape, policymakers are urged to develop real-time systemic risk surveillance frameworks incorporating both dynamic connectedness and causality diagnostics to detect emerging contagion pathways early. Enhancing market liquidity and regulatory clarity for renewable energy and carbon-linked assets is critical, as these markets grow in systemic relevance yet remain susceptible to shocks. Investors would benefit from adopting dynamic, multi-asset allocation strategies that adjust to the temporal variability of risk transmission, supported by advanced econometric tools such as DCC-GARCH and TVGC. Strong coordination between financial regulators, energy authorities, and environmental policymakers will be vital to managing cross-sector vulnerabilities and supporting the ongoing global energy transition sustainably. Promoting transparent disclosure of volatility, interconnectedness metrics, and risk exposures will empower all stakeholders to anticipate market stress scenarios and make informed decisions.

Together, these integrated findings from decomposed connectedness and predictive causality analyses provide a comprehensive understanding of the structural and temporal risk dynamics in international energy markets, informing robust investment strategies and effective policymaking aligned with the challenges and opportunities of the clean energy transition.

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